

Preparing to study the surface of Phobos using IDEFIX WheelCam Images

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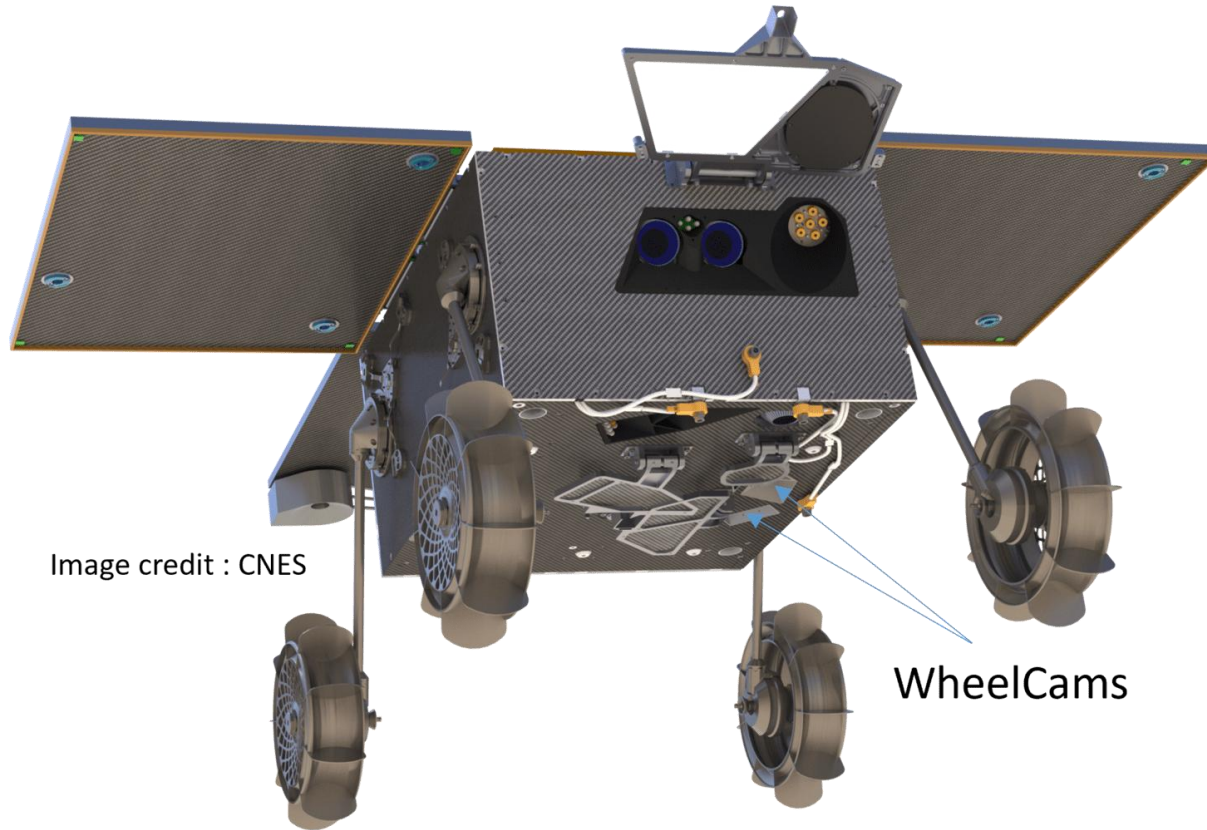
INDEX

- Introduction
- Theoretical Background
- Wheel Sinkage: Methodology and results.
- Spectroscopy: Methodology and results.
- Conclusions and future work



MMX MISSION OBJECTIVES

- Clarify the origins of the Martian moons
- Characterize surface and regolith properties of Phobos



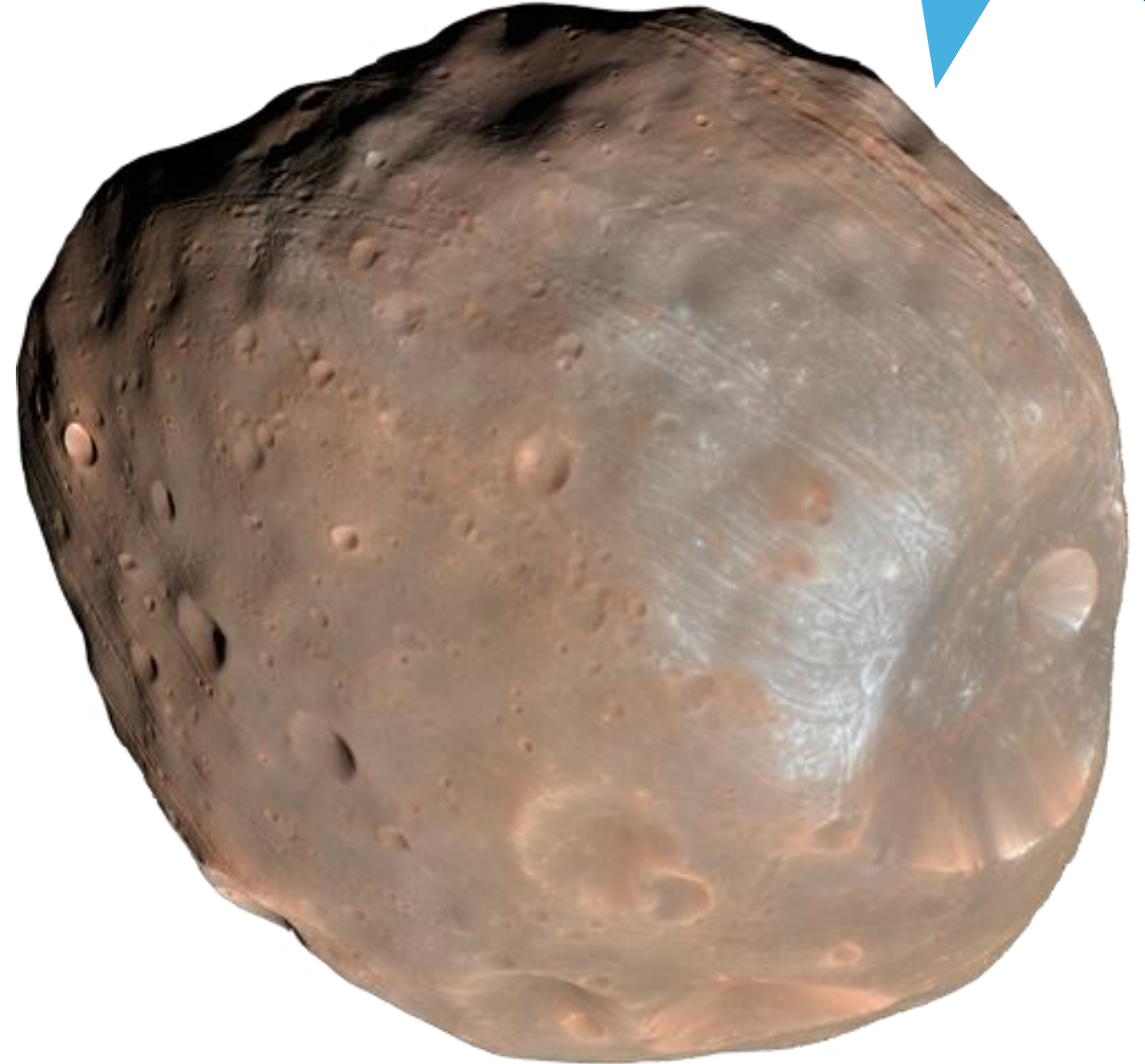
IDEFIX OBJECTIVES

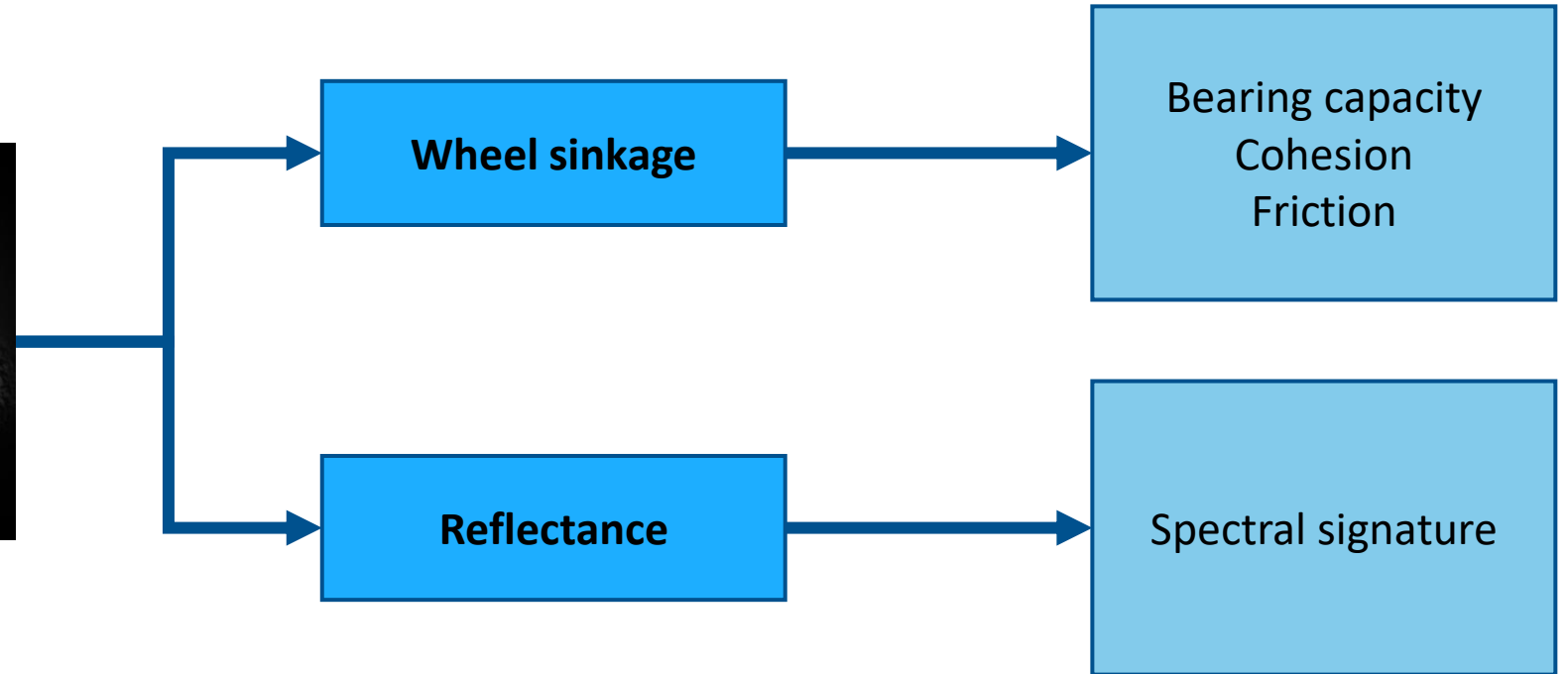
- Characterize the composition and heterogeneity of Phobos' Regolith
- Investigate the physical and mechanical properties of the regolith through rover-surface interactions



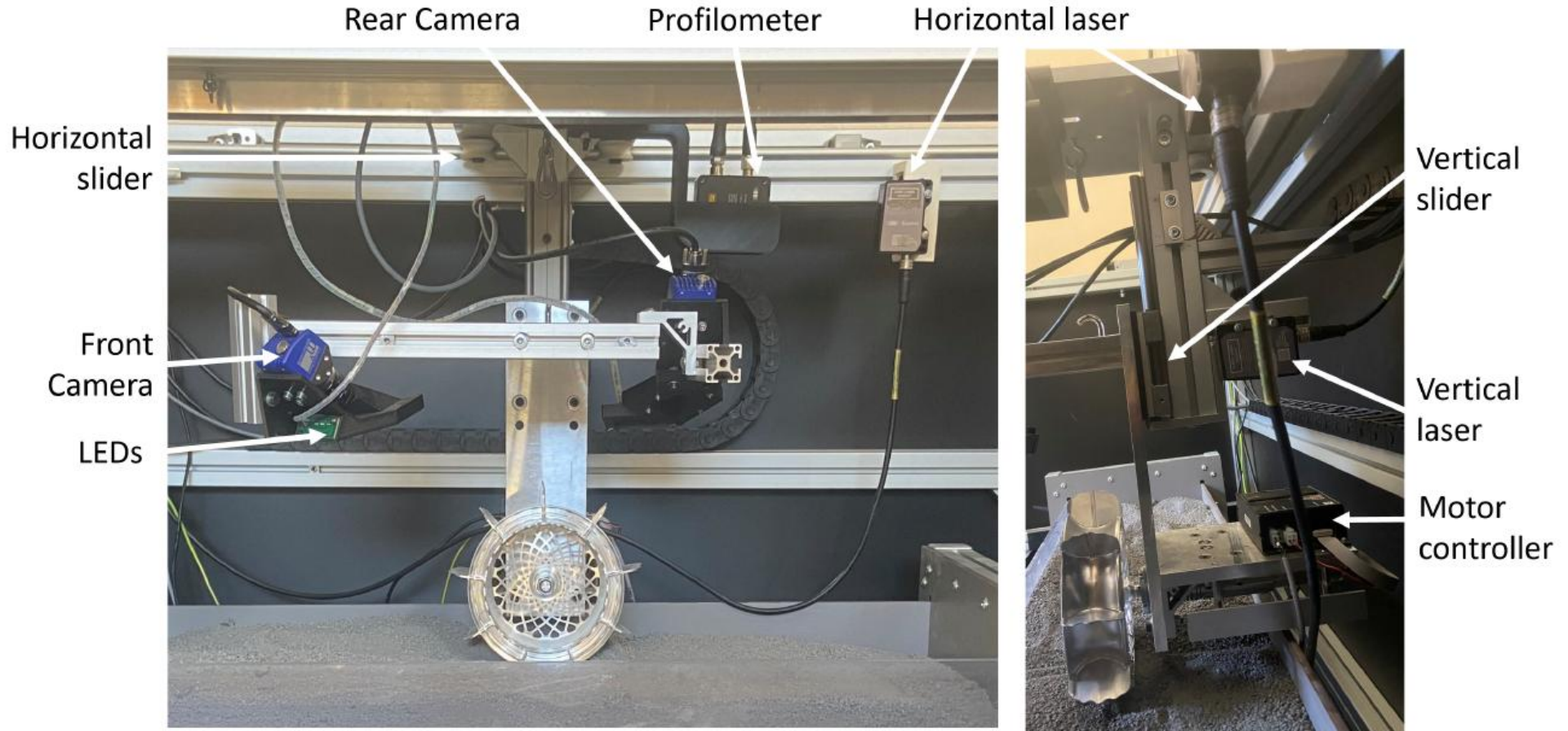
OBJECTIVES

- Develop an algorithm that can isolate the wheel and estimate the sinkage
- Determine how the combination of the colored LEDs and the panchromatic camera can provide information about the surface of Phobos.





EXPERIMENTAL TESTBED



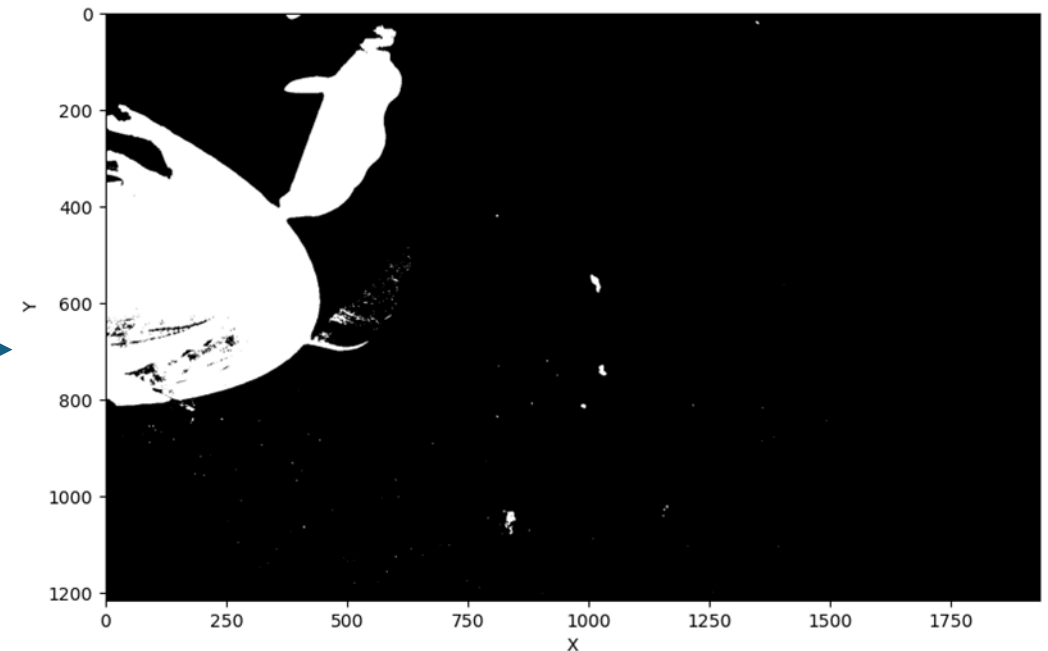
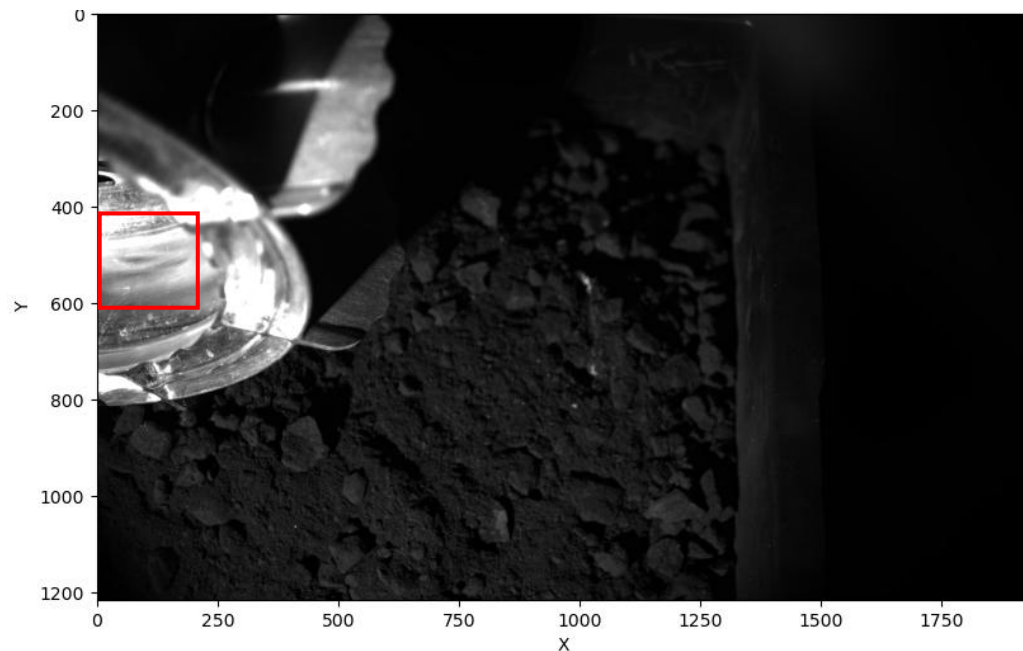


WHEEL SINKAGE

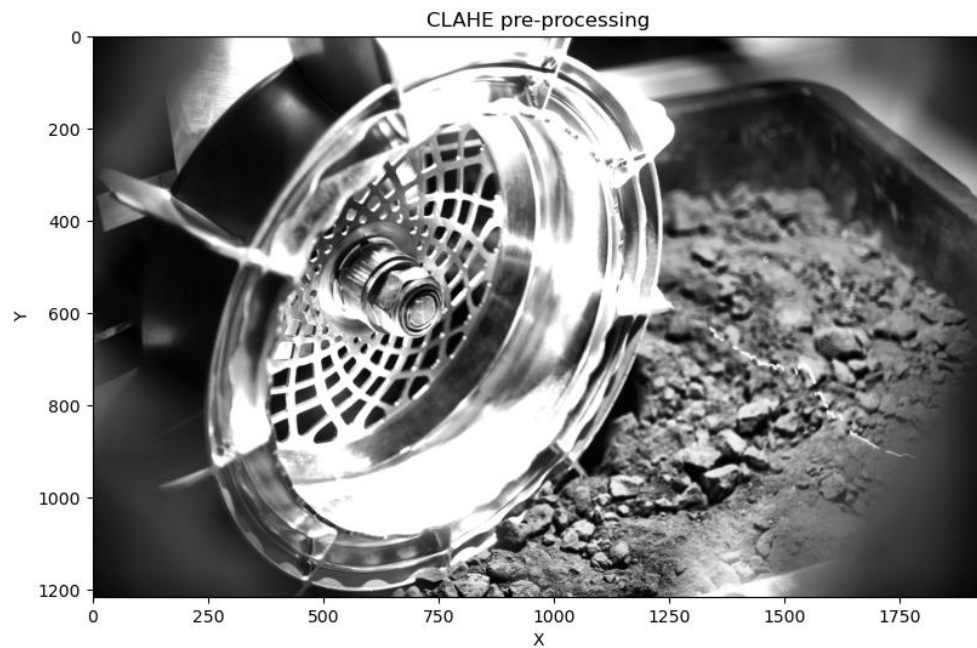
METHODOLOGY AND RESULTS

- Segmentation
- Morphological transformations `cv2.MORPH_OPEN`
`cv2.MORPH_CLOSE`

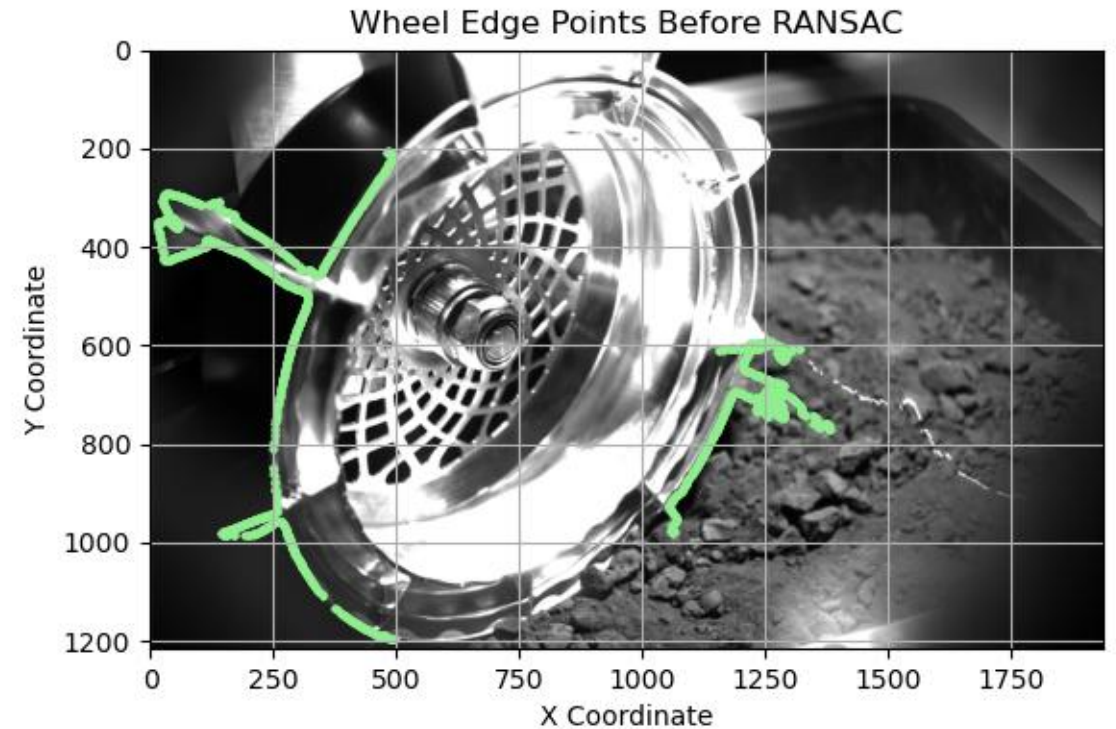
- Edge detection `cv2.GaussianBlur()`
`cv2.Canny()`
- Contour selection `cv2.findContours()`
`cv2.minEnclosingCircle()`



- CLAHE preprocessing



- Piecewise image segmentation



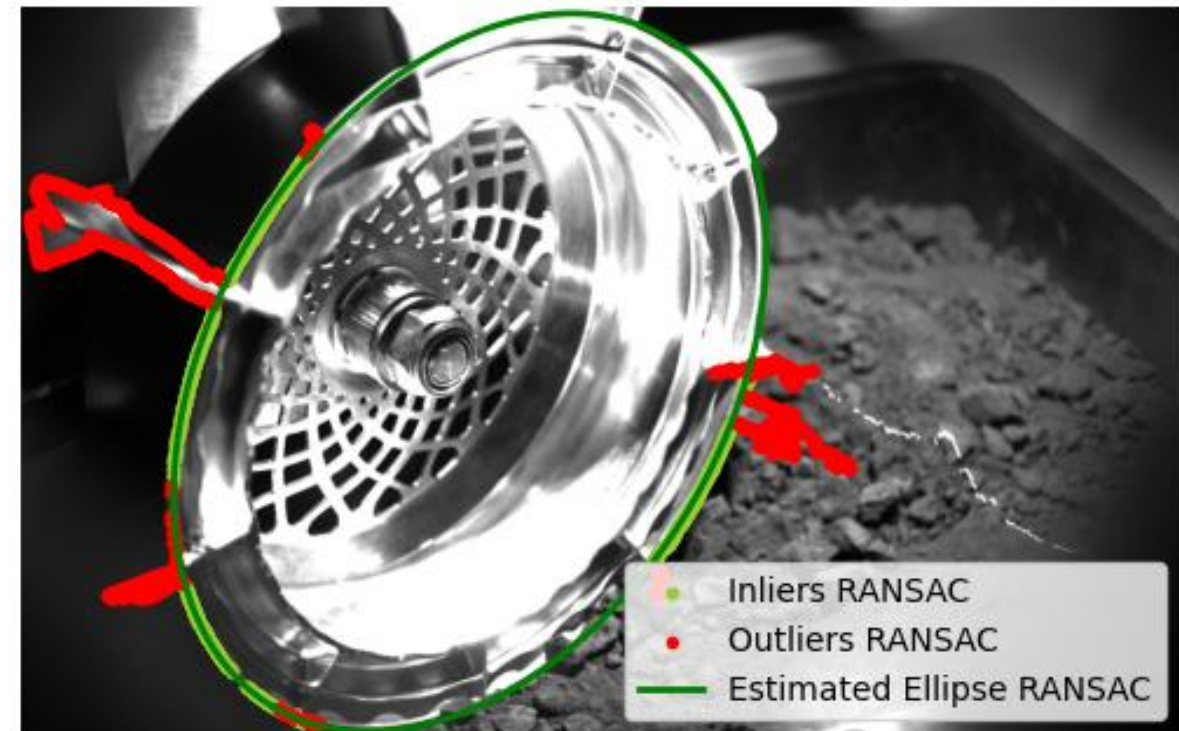
RANSAC based ellipse fitting

- Inlier to outlier ratio
- Spatial dispersion of points



Quantifying ellipse fit accuracy

- $d_H(A, B) = \max \left\{ \sup_{a \in A} \inf_{b \in B} |a - b|, \sup_{b \in B} \inf_{a \in A} |b - a| \right\}$
- $IoU = \frac{A \cap B}{A \cup B}$



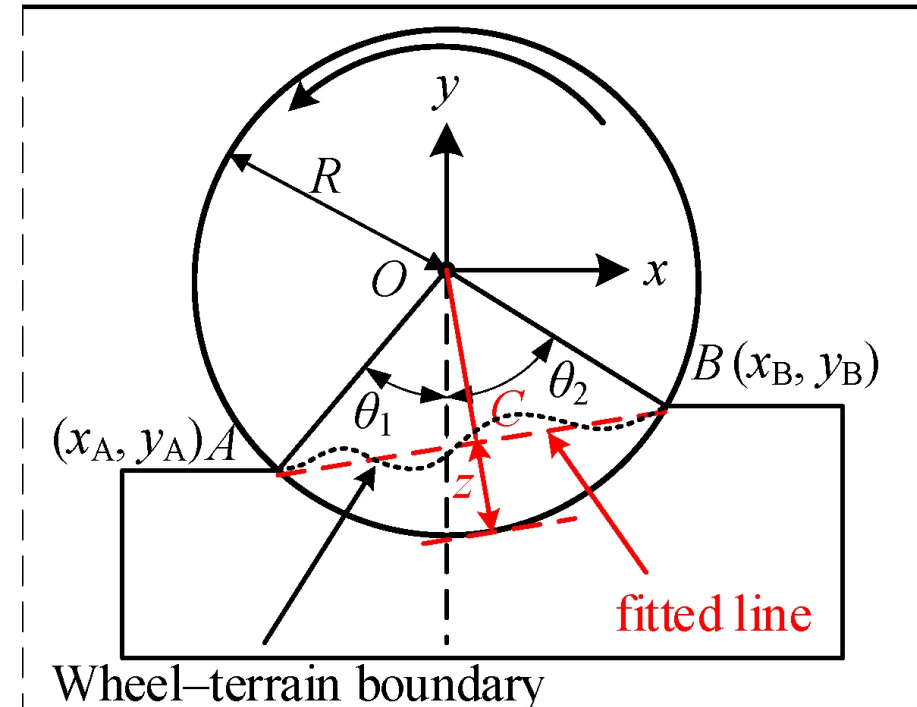
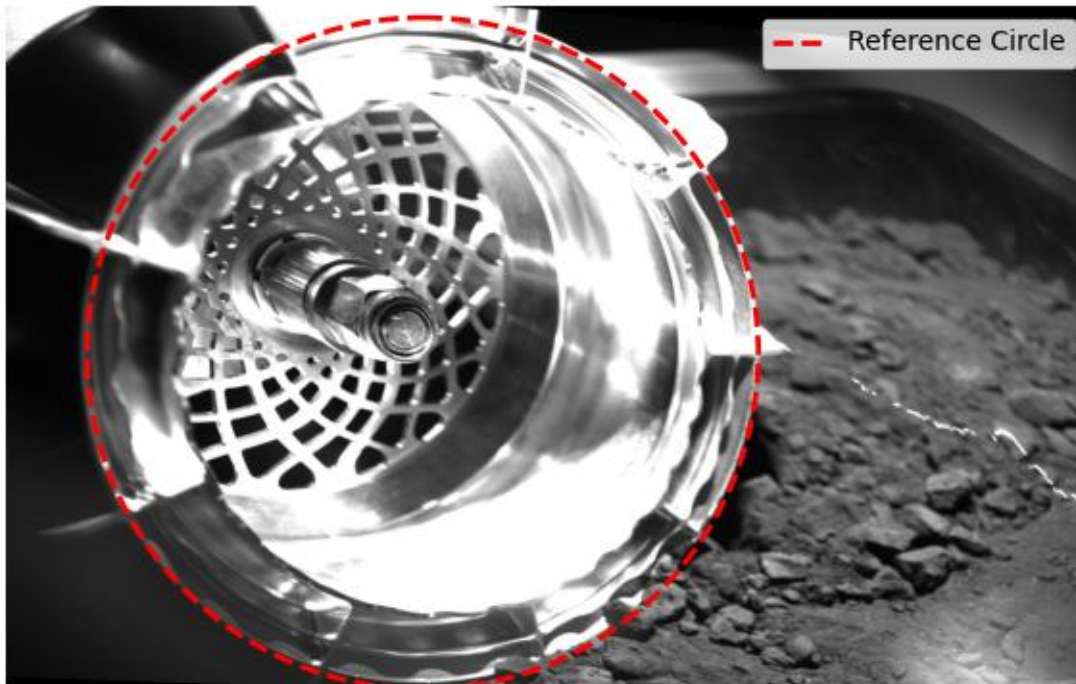
Homography transform

- Normalise geometry with respect to the camera angle

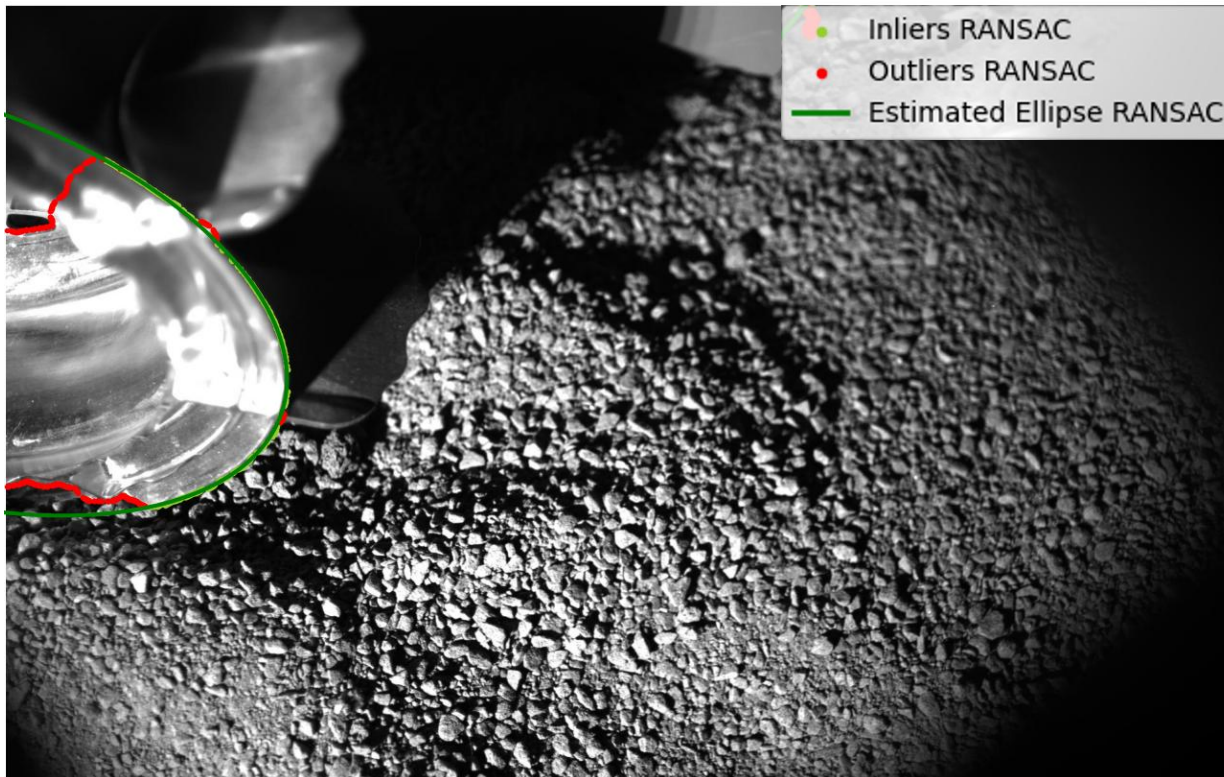
Sinkage estimation

- Wheel-regolith interface required

Transformed Image (ellipse->circle)

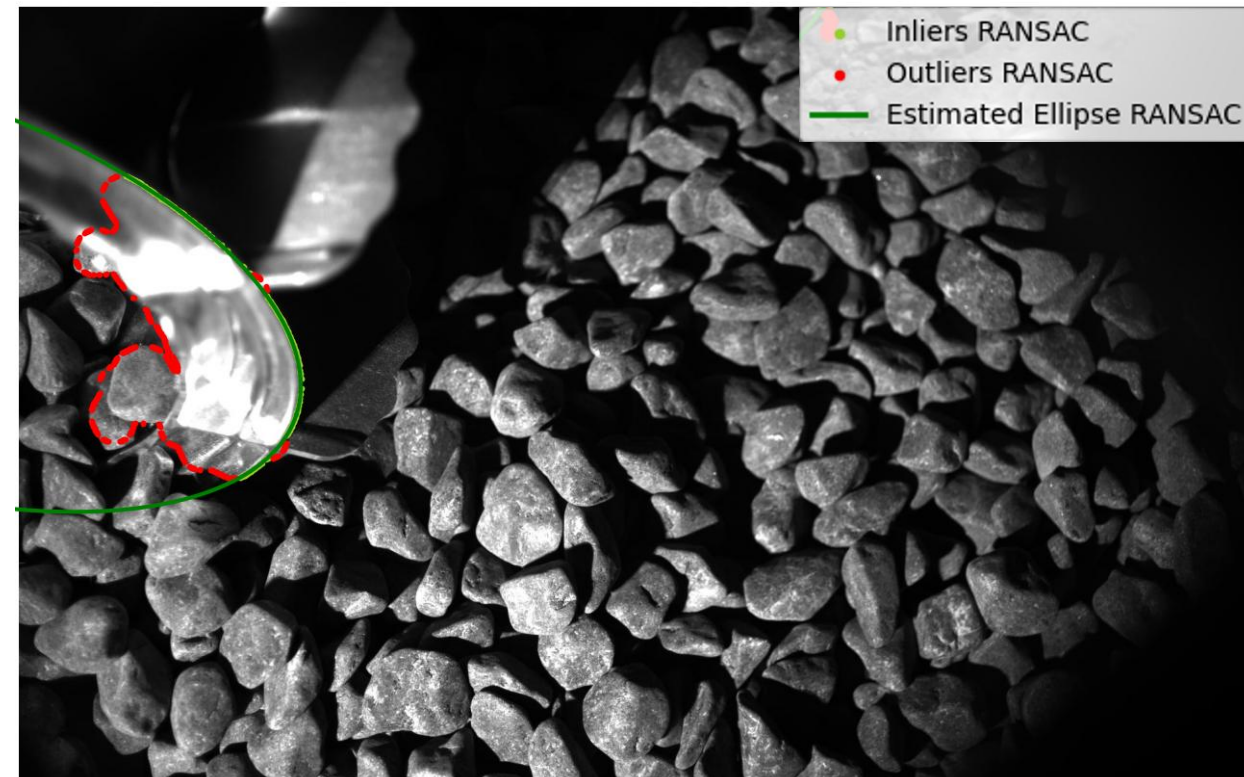


Moon simulant, low sinkage



- 656 inliers, 461 outliers

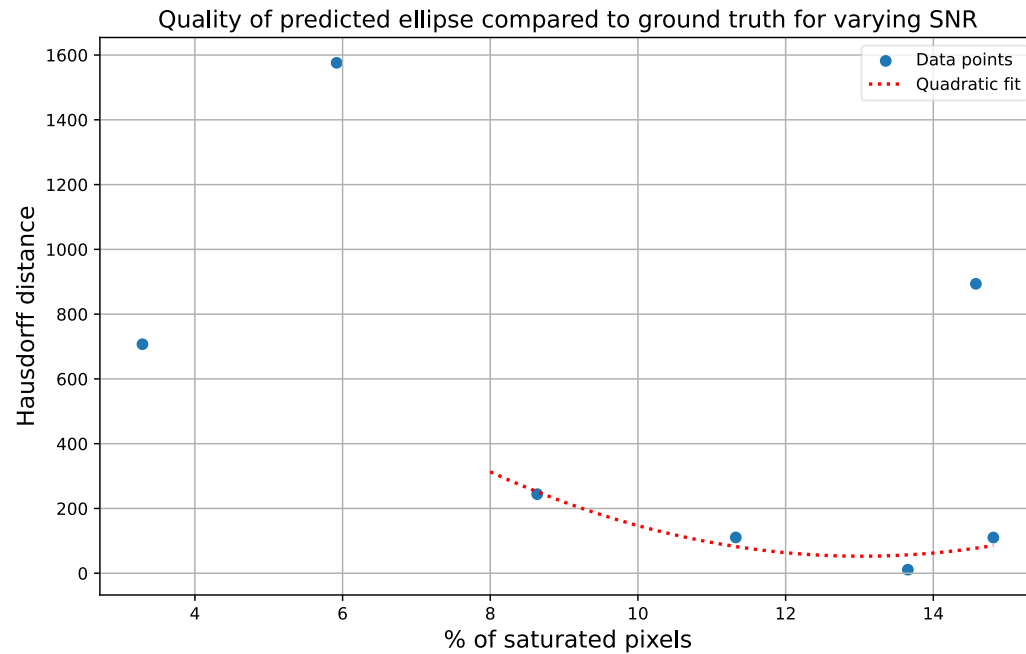
Coarse gravel, high sinkage



- 492 inliers, 851 outliers

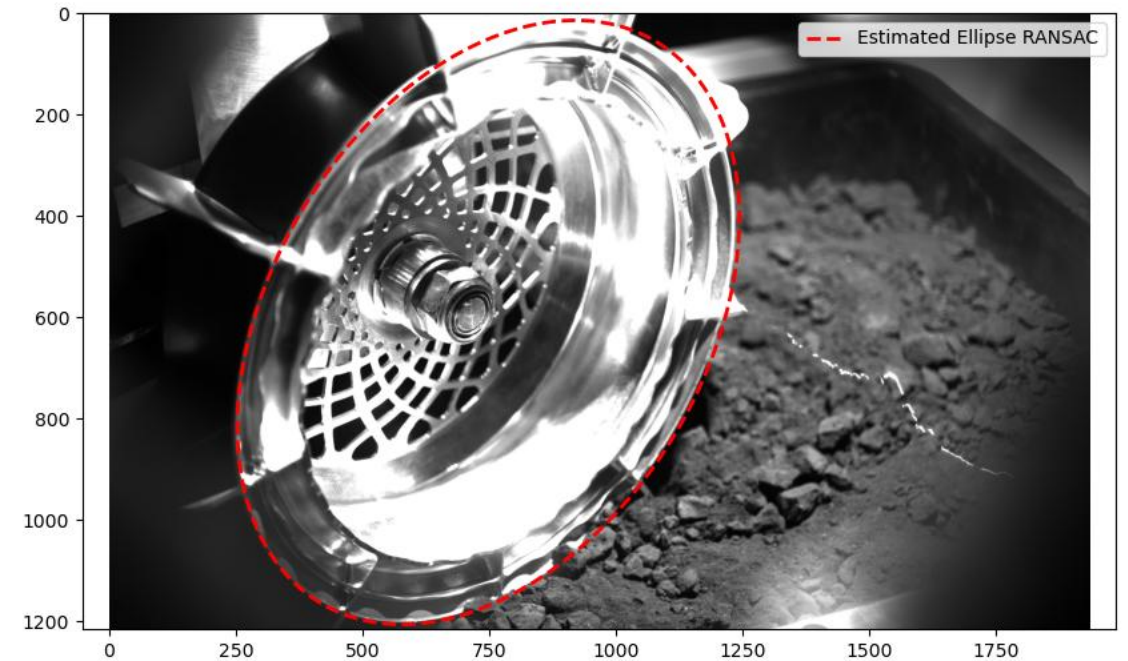
Front WheelCam – Sensitivity to illumination conditions and regolith simulants

- Effect of saturation on ellipse fitting quality
- Experiment performed on JAXA and Moon simulants



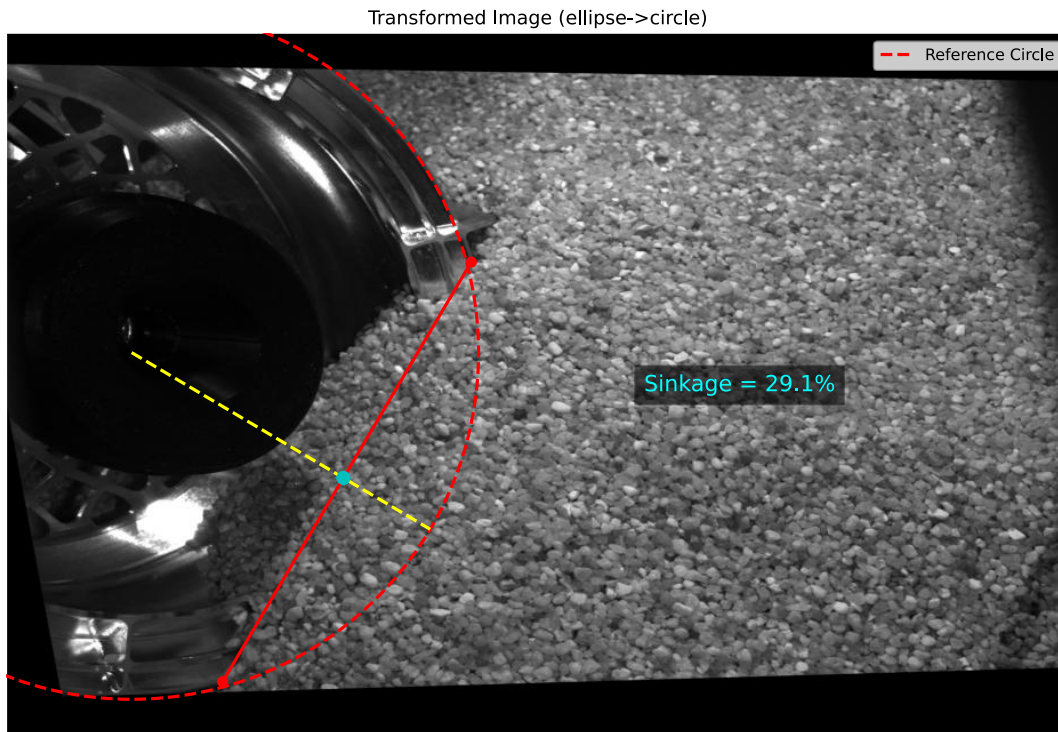
Ideal saturation point

- JAXA simulant 13.6%
- Moon simulant 3.1%



Ellipse fit at 13.6% saturation, JAXA regolith simulant

- 1) Relate ellipse fit metrics and sinkage estimation accuracy
- 2) Validate sinkage estimation with laser measurements



Automatic vs. manual ellipse fit

d_H	257
IoU (%)	74
Absolute error (mm)	2.52
Relative error (%)	11.07

Comparison between automatic ellipse and manual ellipse

Automatic vs. laser measurement

Sinkage (% R_w)	29.10	33.74
Absolute error (mm)	3.86	
Relative error (%)	13.75	

Comparison between automatic ellipse and laser measurement

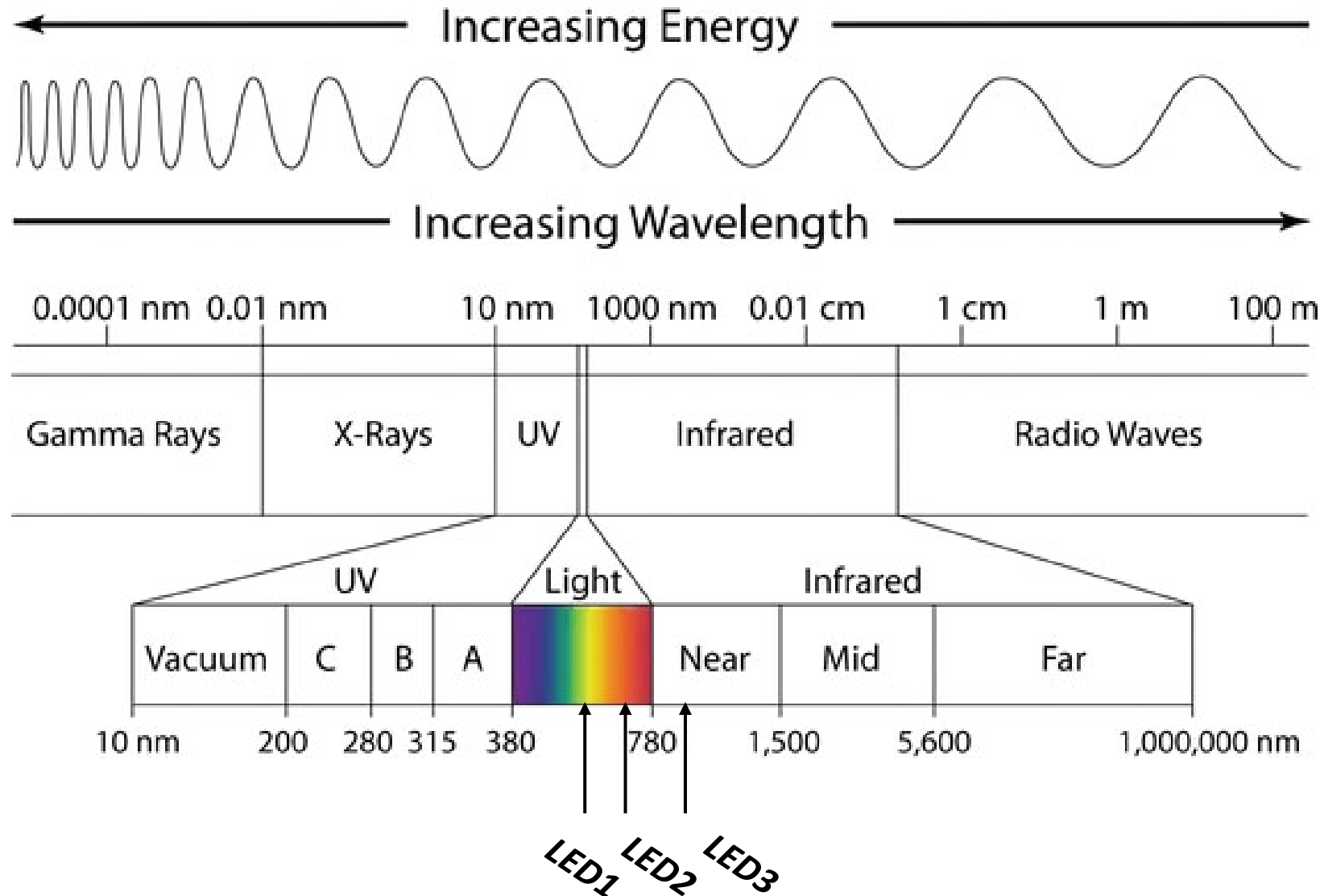


SPECTROMETRY

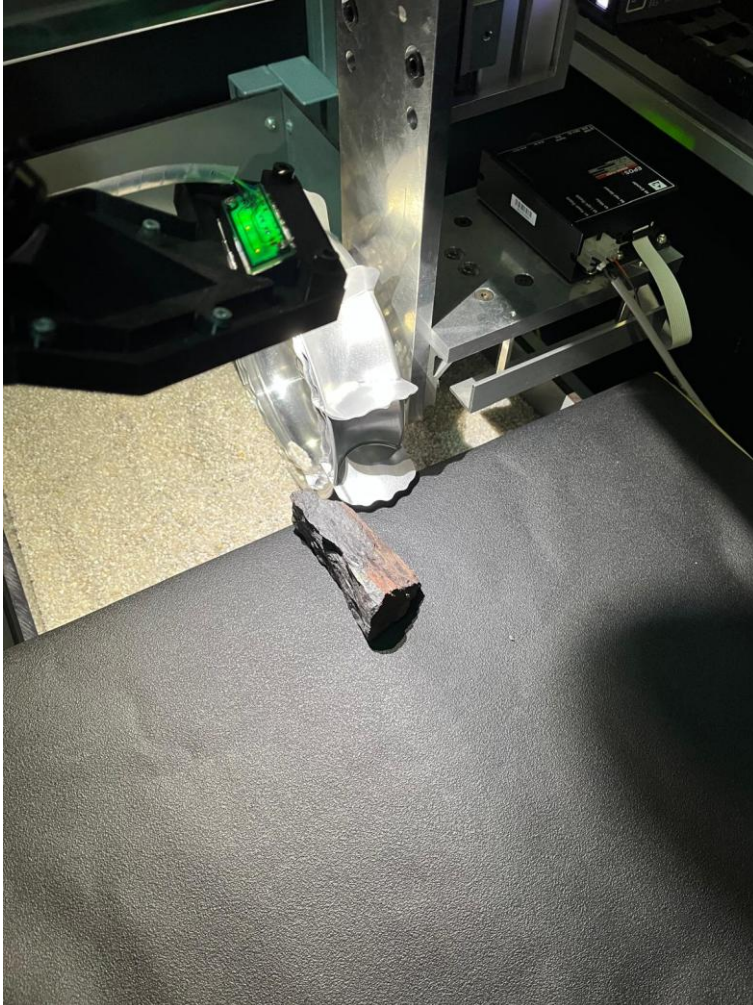
METHODOLOGY AND RESULTS

METHODOLOGY

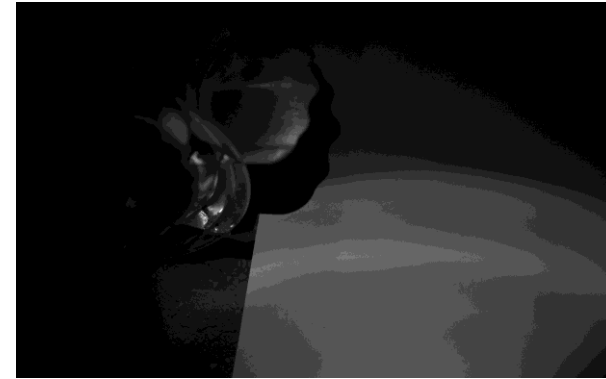
SPECTROMETRY



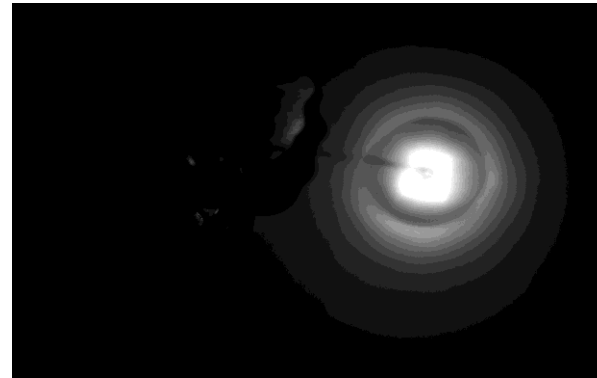
SETUP



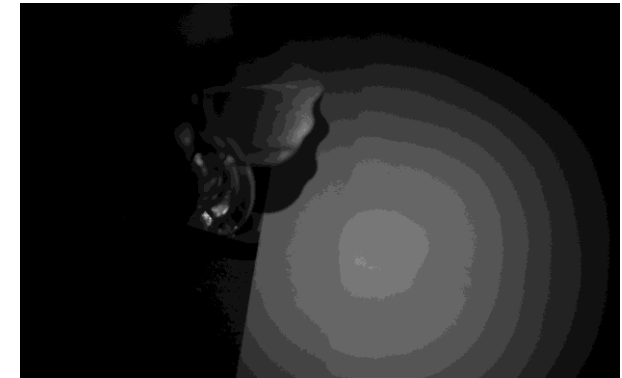
LED1 (590nm Yellow)



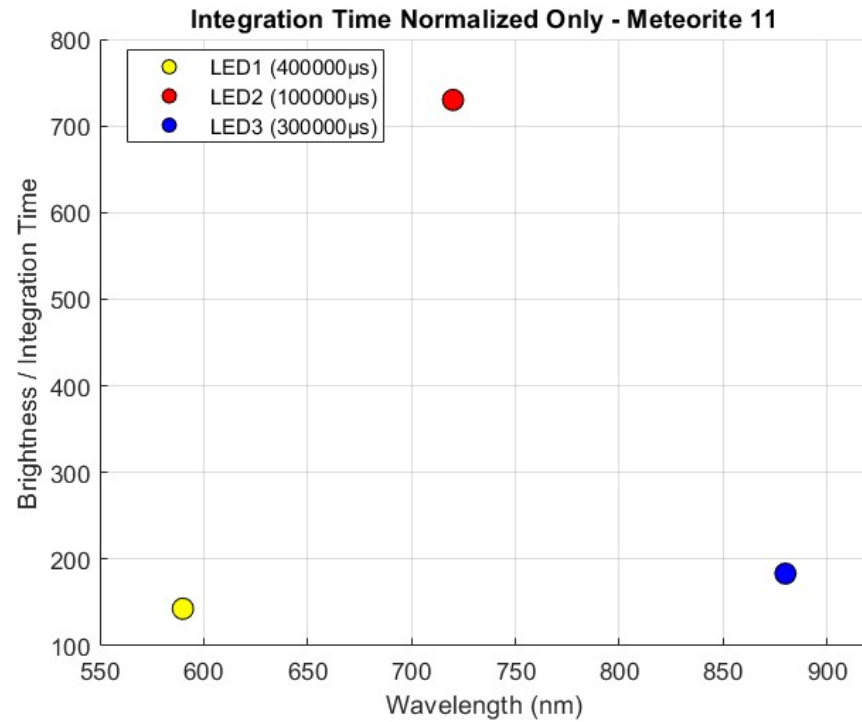
LED2 (720 nm Red)



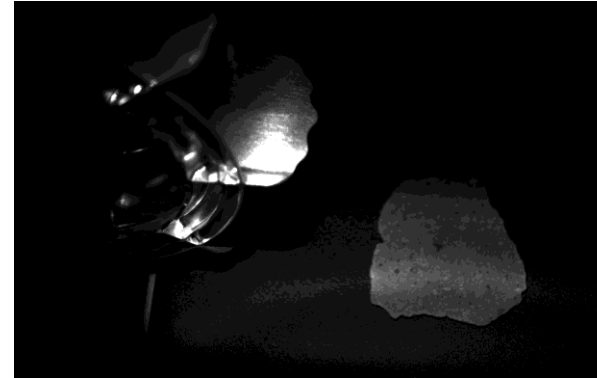
LED3 (880 nm IR)



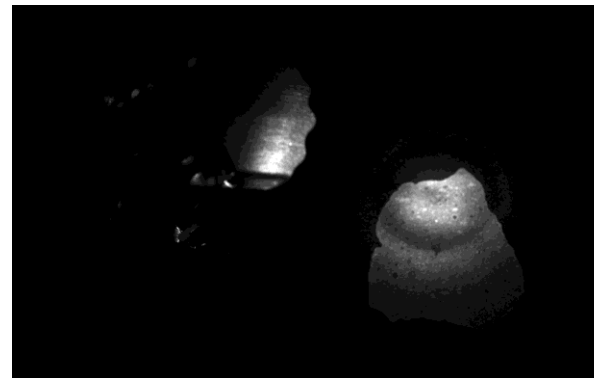
SETUP



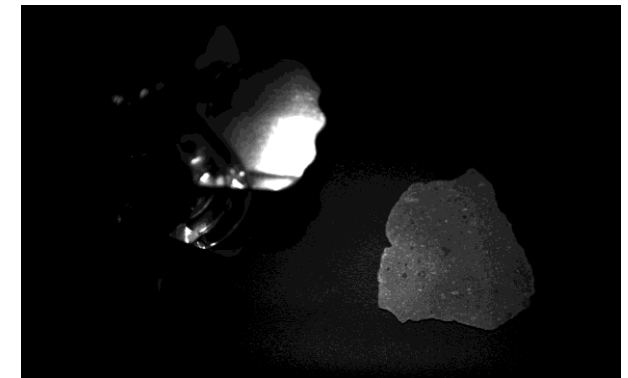
LED1 (590nm Yellow)



LED2 (720 nm Red)



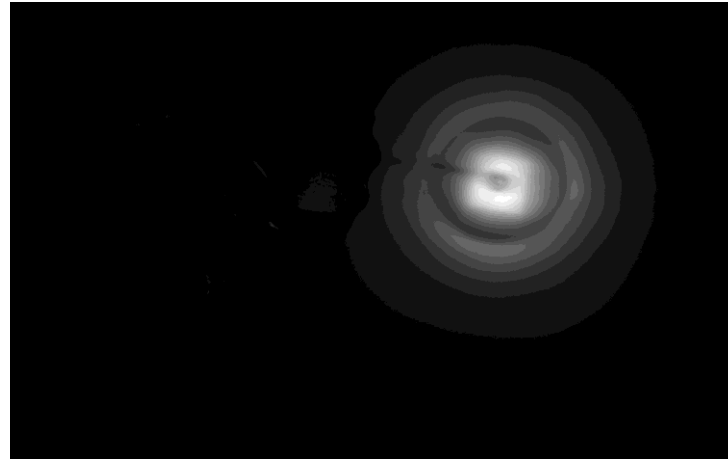
LED3 (880 nm IR)



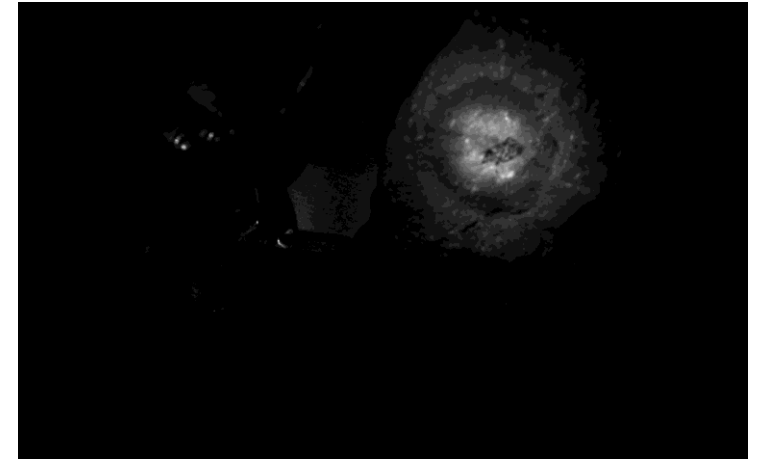
$$R_s = 0.95$$



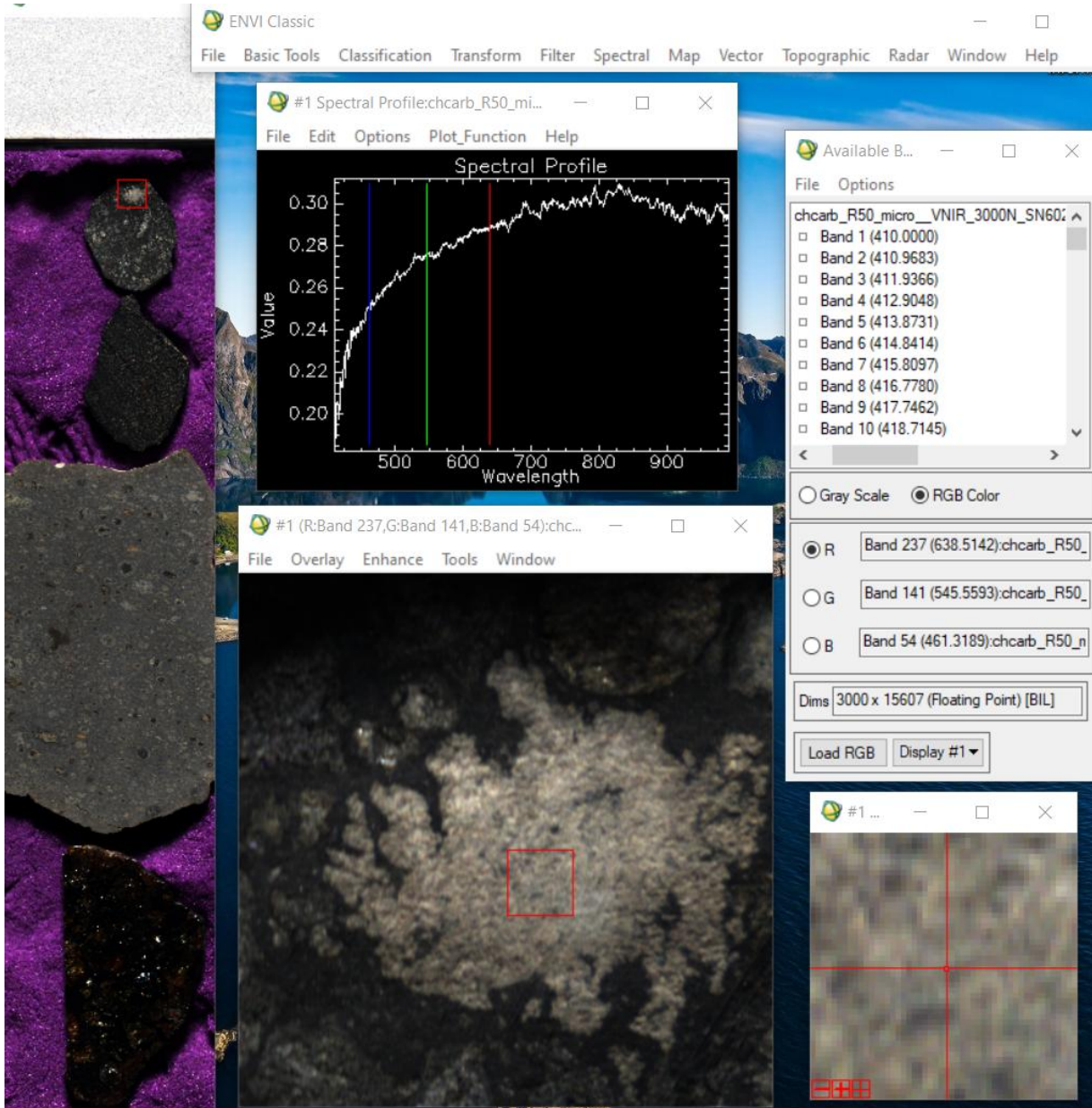
LED2 on calibration panel



LED2 on sample



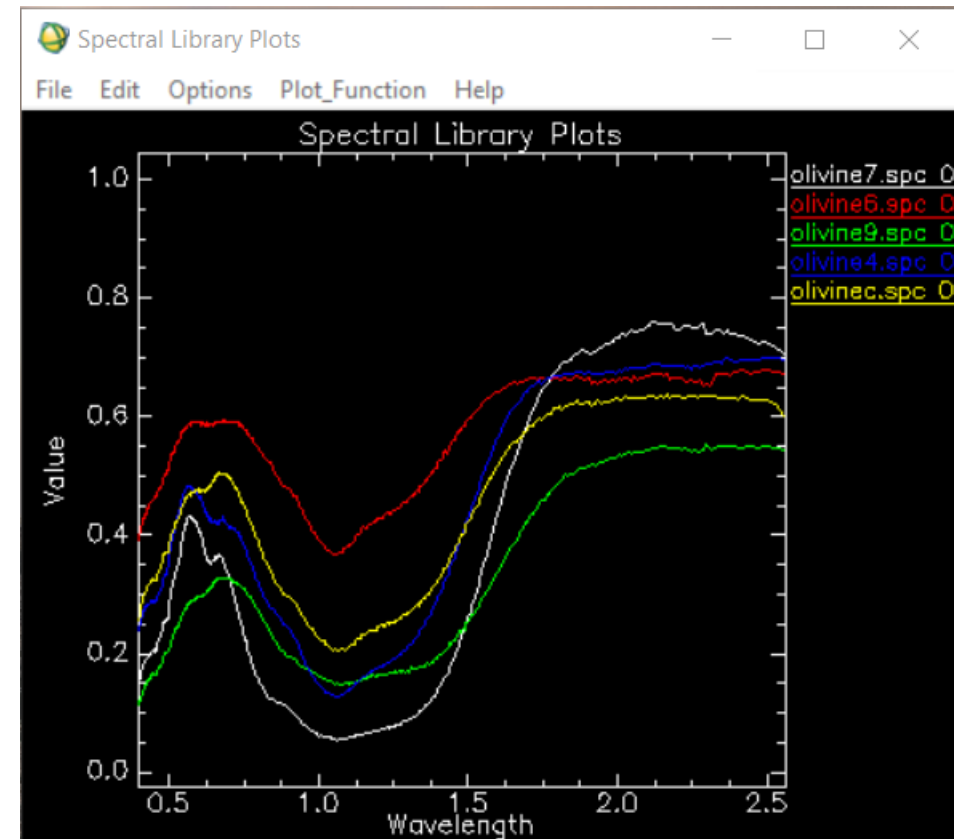
$$B_c(\lambda) = L_{LED}(\lambda) \times R_c(\lambda) \times t_c \quad B_s(\lambda) = L_{LED}(\lambda) \times R_s(\lambda) \times t_s$$



Spectral Cube

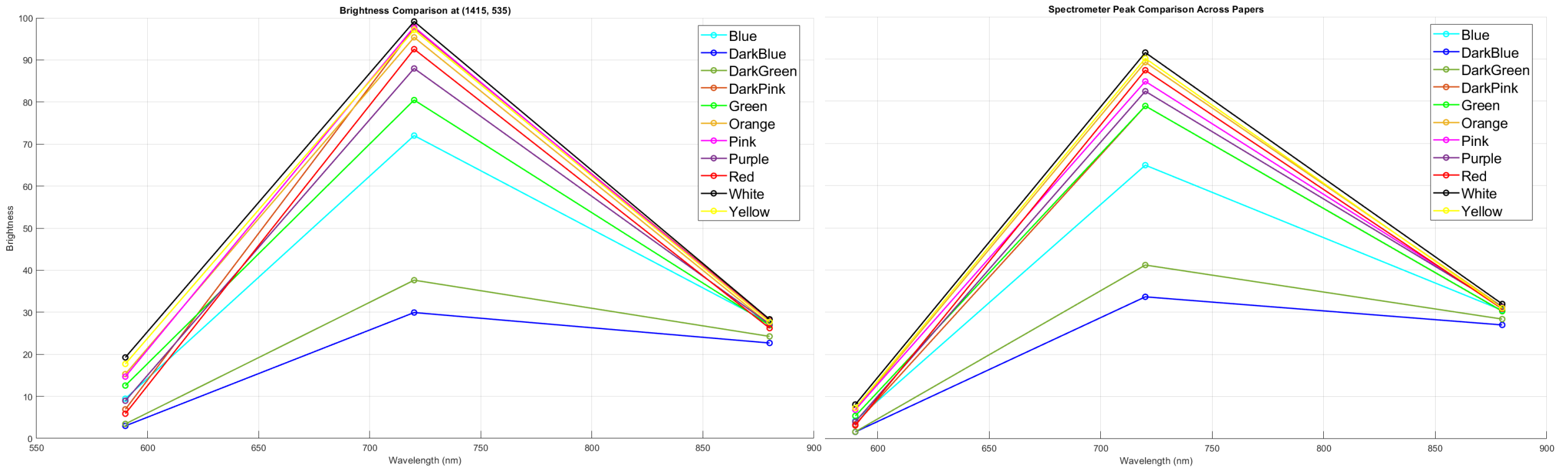
METHODOLOGY

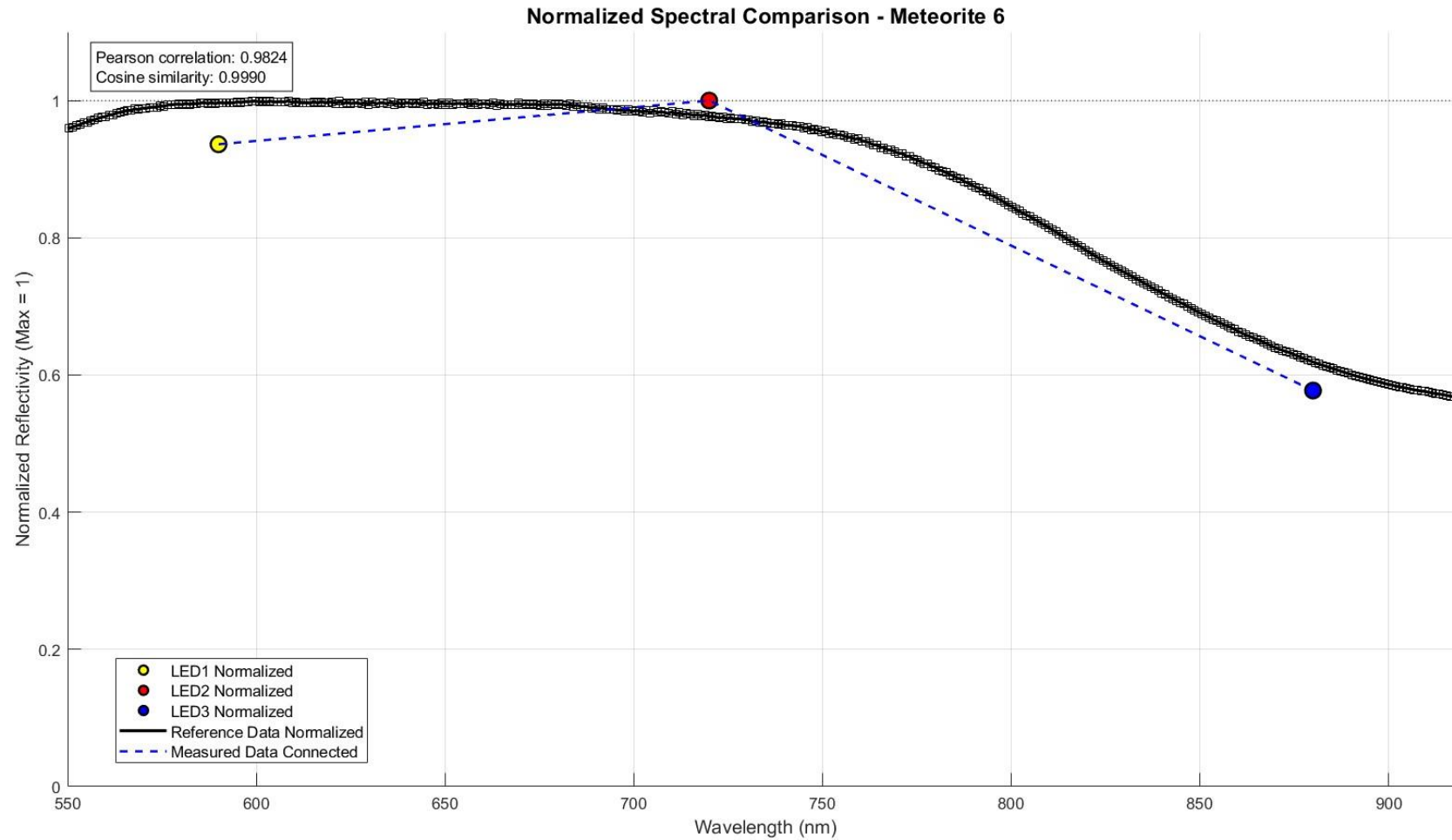
REFERENCE DATA

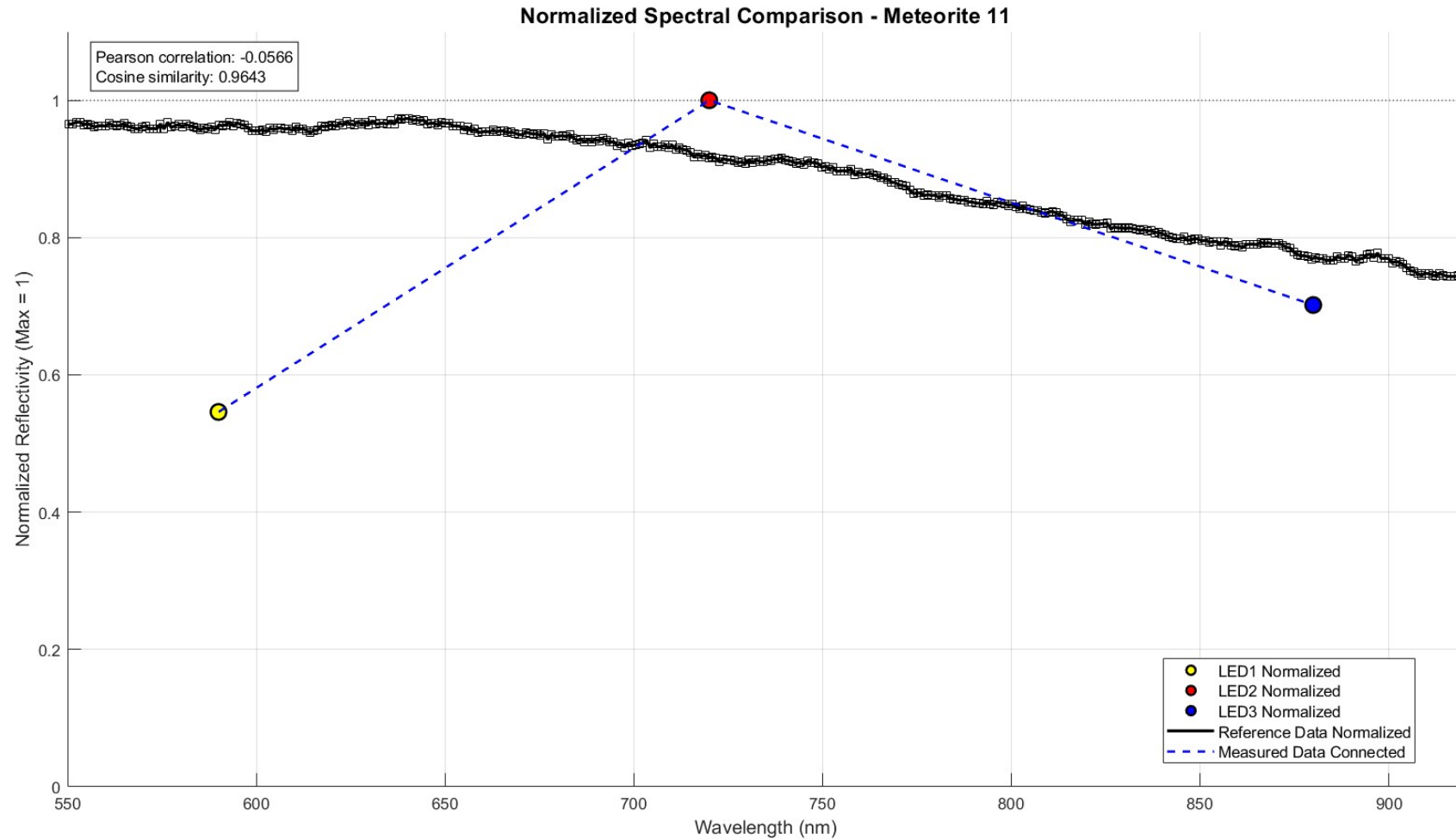


Spectral Library

- Promising results -> geological samples
- Important findings: misalignment of LEDs, intensity difference -> need for calibration



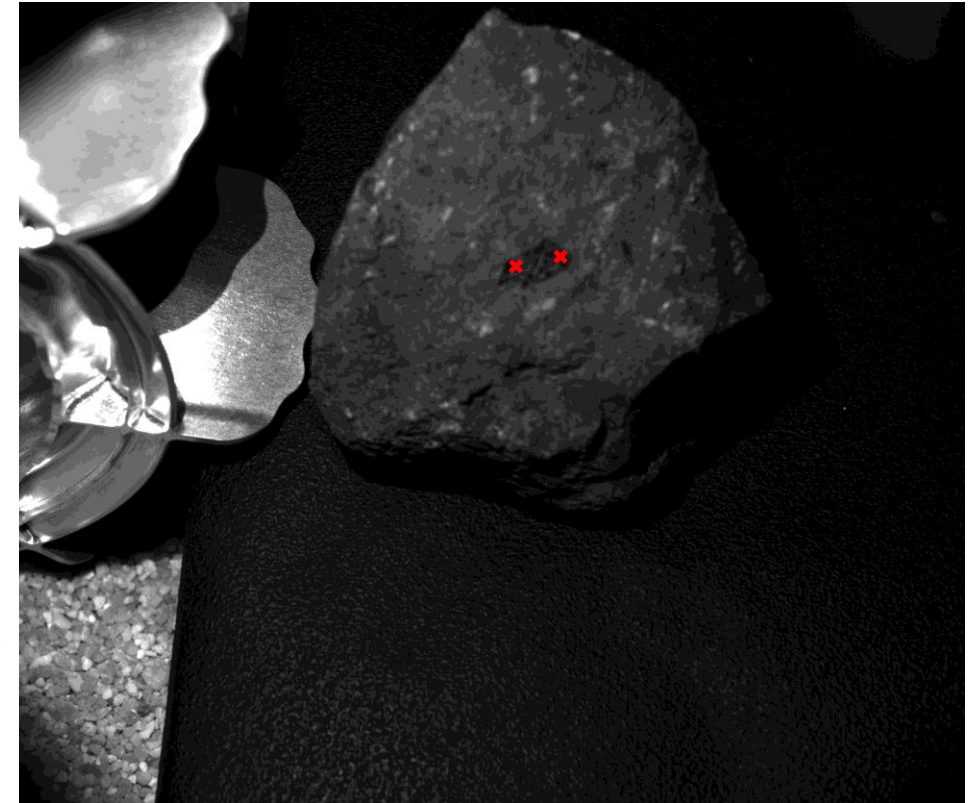
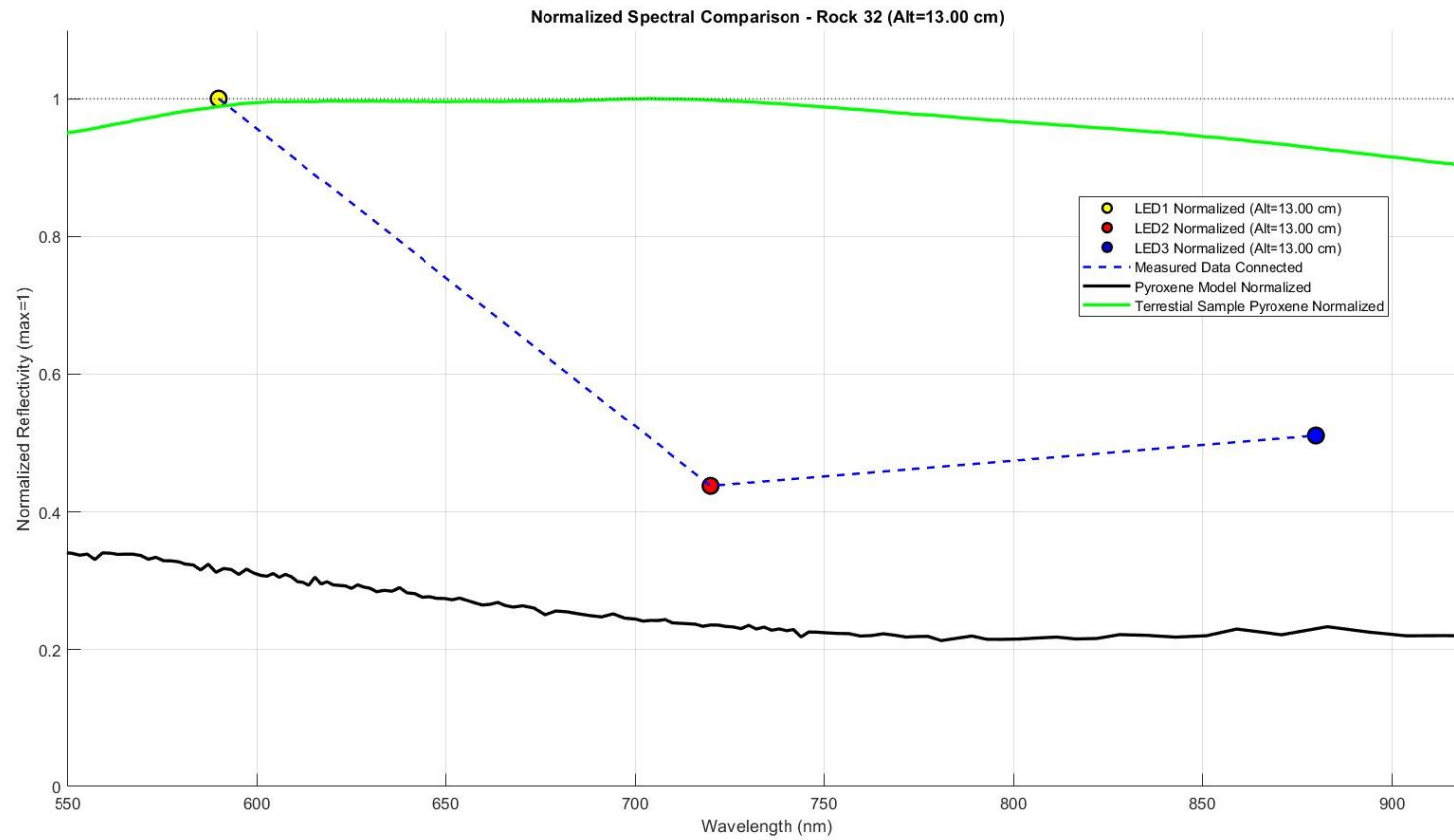




RESULTS

ANALOGUE SAMPLES





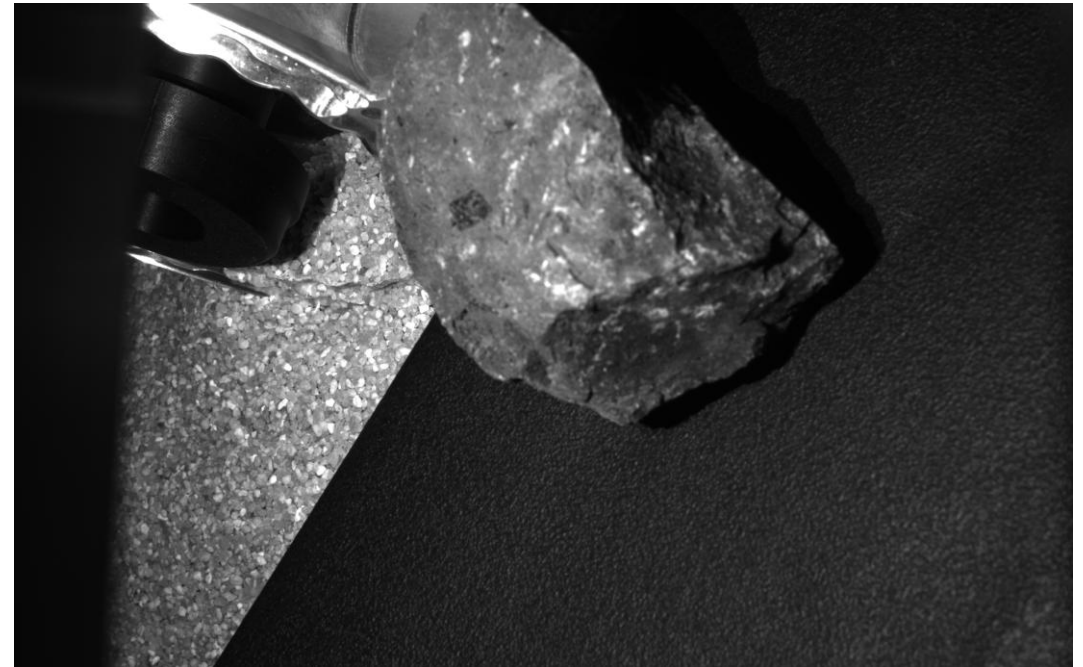
RESULTS

ANALOGUE SAMPLES

Previous configuration



Updated configuration





WHEEL DETECTION

- Robust contour detection and ellipse fitting across regolith simulants
- Ellipse fitting accuracy depends on LED illumination and regolith reflectivity
- Sinkage estimation tested against laser measurements, <5% deviation observed

SPECTROMETRY

- Better results for the meteorites because of the spectral cube
- Method shows sensitivity to composition and texture
- Results highly dependent on: point selection, height, angle and surface texture
- Not yet reliable for mineral identification

WHEEL DETECTION

- Optimize algorithm for the revised testbed setup
- Expand sinkage validation dataset
- Fully automate the pipeline with a regolith segmentation model



SPECTROMETRY

- Return to spectrometer use
- Study influence of sample texture
- LED characterization with calibrating panel
- Investigate the use of the LEDs for other applications.



- Murdoch, N. et al (2025). *The WheelCams on the IDEFIX rover*.
- *MMX Exploration Mission*. [Online].
Available: https://www.esa.int/Science_Exploration/Space_Science/MMX_factsheet
- *The Radio Spectrum: Understanding ITU Frequency Bands from VLF to UHF*. [Online]. Available: <https://www.ariat-tech.com/blog/the-radio-spectrum-understanding-itu-frequency-bands-from-vlf-to-uhf.html>
- *Phobos*. [Online].
Available: <https://thesolarsystem.fandom.com/wiki/Phobos>

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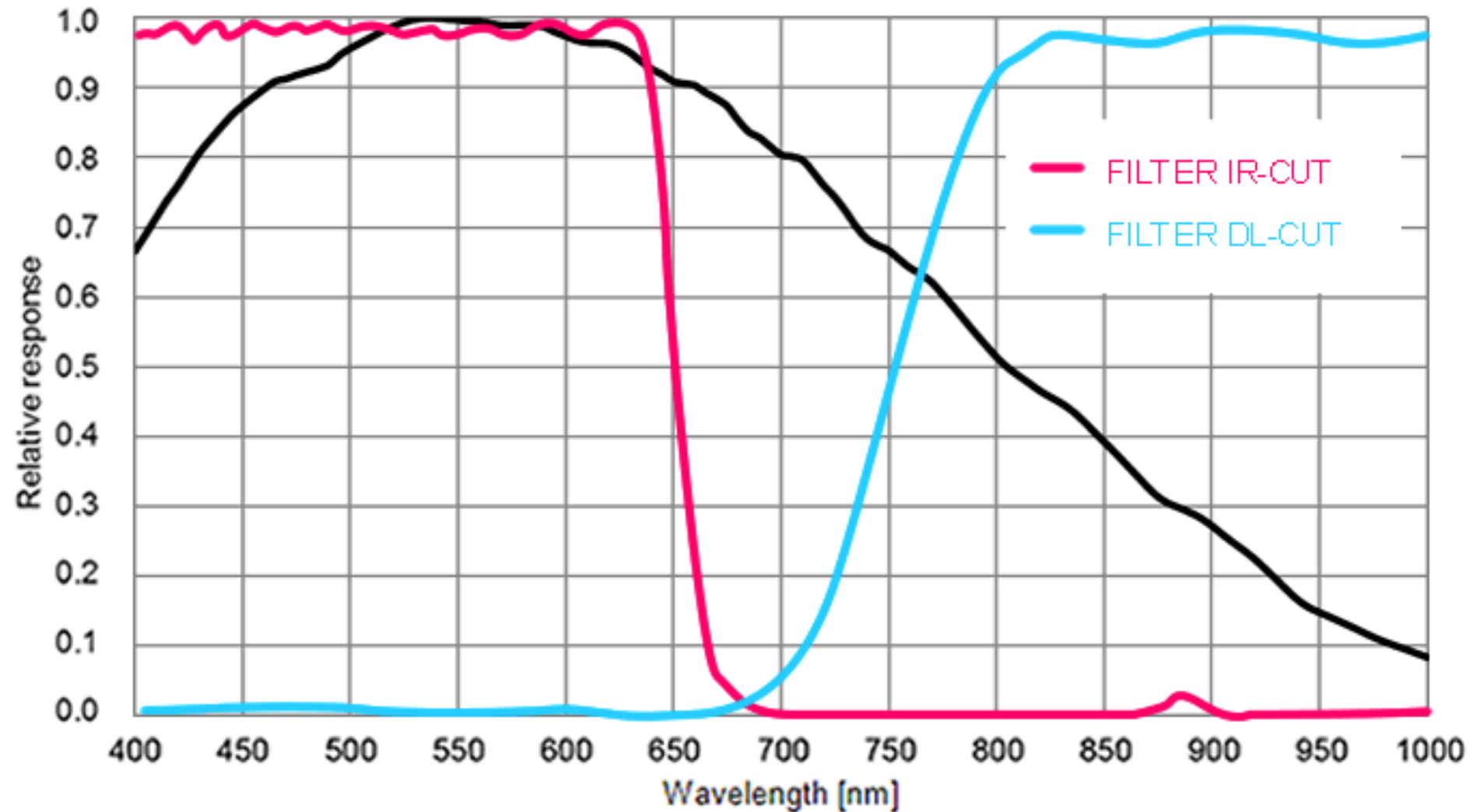
Dr. Naomi Murdoch

Alice Amsili



- Key parameters: Cohesion, internal friction angle, shear strength and stiffness
- Modeling approach: Semi-empirical terramechanics models (Bekker and Reece-Wong models)
- Important to adapt models to low-gravity conditions
- Fit sinkage and slip data
- Bearing capacity: independently from rover trench morphology and sinkage analysis

Camera sensitivity



- Need lots of training data
- Adaptation to unseen conditions is unpredictable
- Could be a solution to automate wheel regolith interface detection



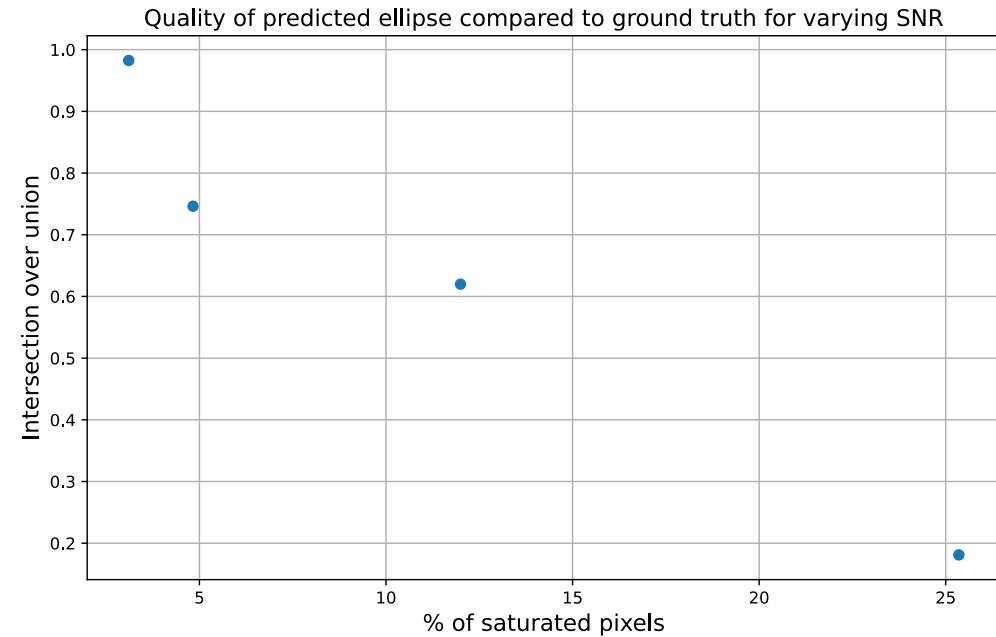
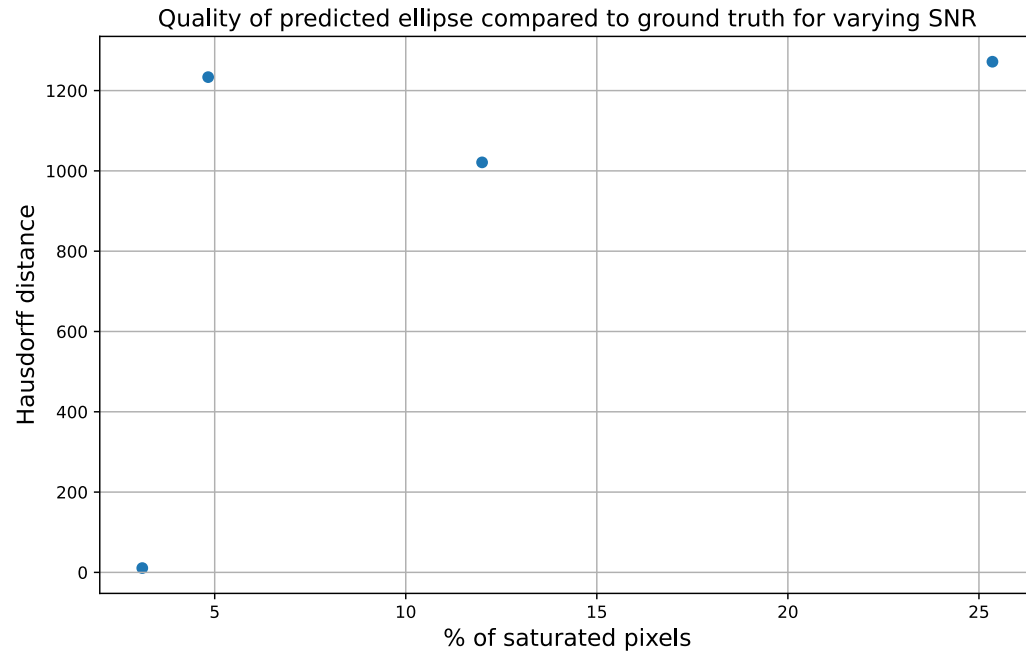
ML wheel segmentation model

Ellipse fit under variable SNR

WHEEL SINKAGE

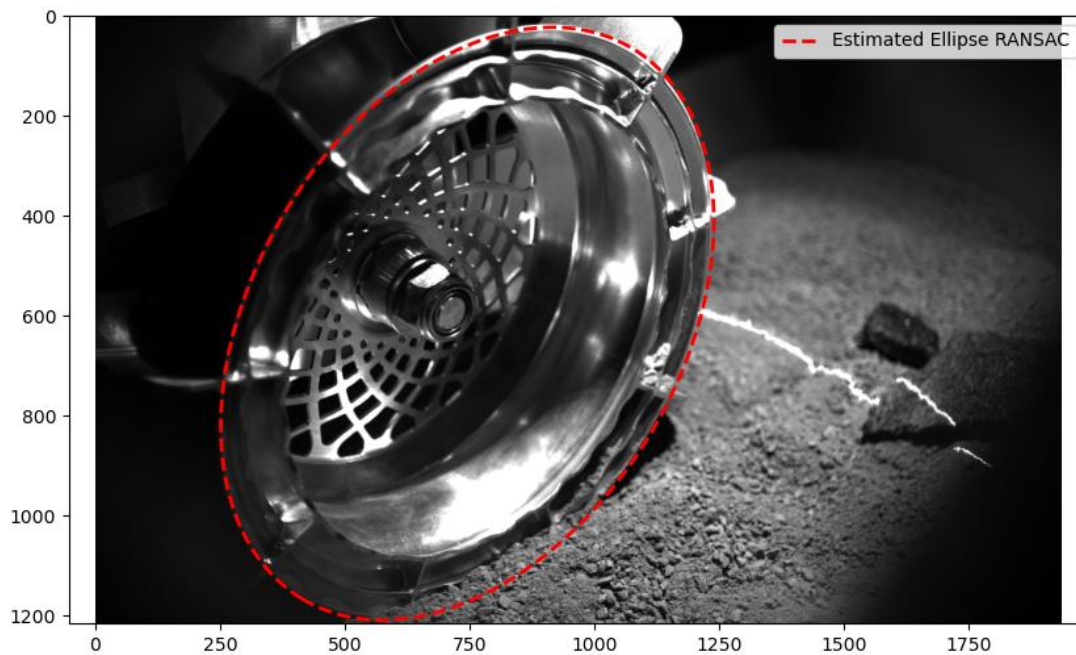


- Moon data



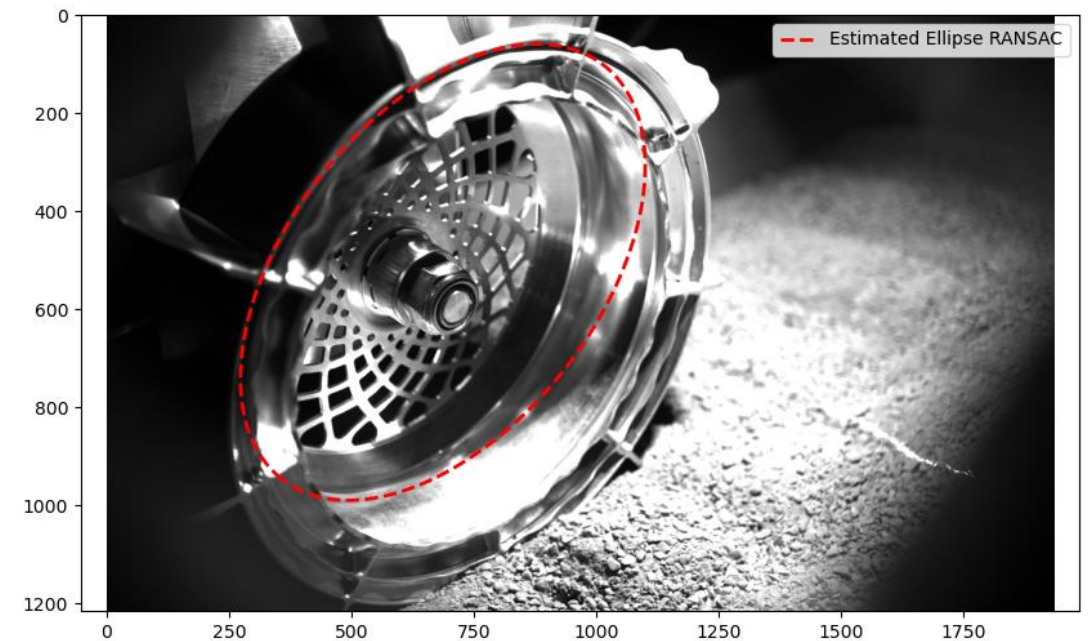


- Moon data



Optimal exposure, Moon regolith simulant (3% saturation)

$$d_H = 10.740, IoU = 0.982$$



Over-exposure, Moon regolith simulant (12% saturation)

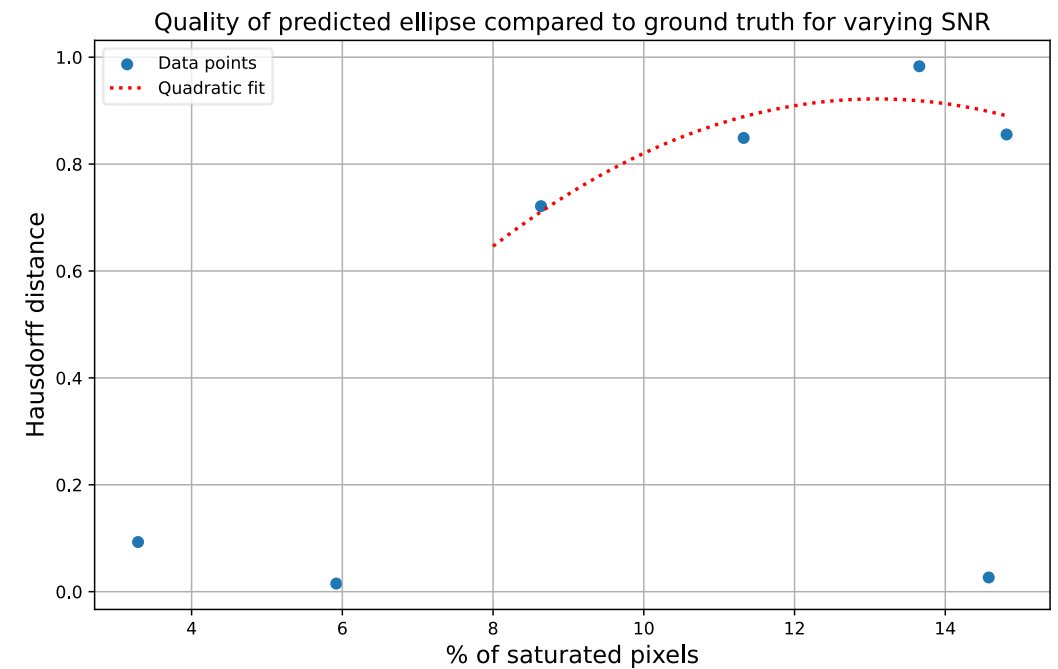
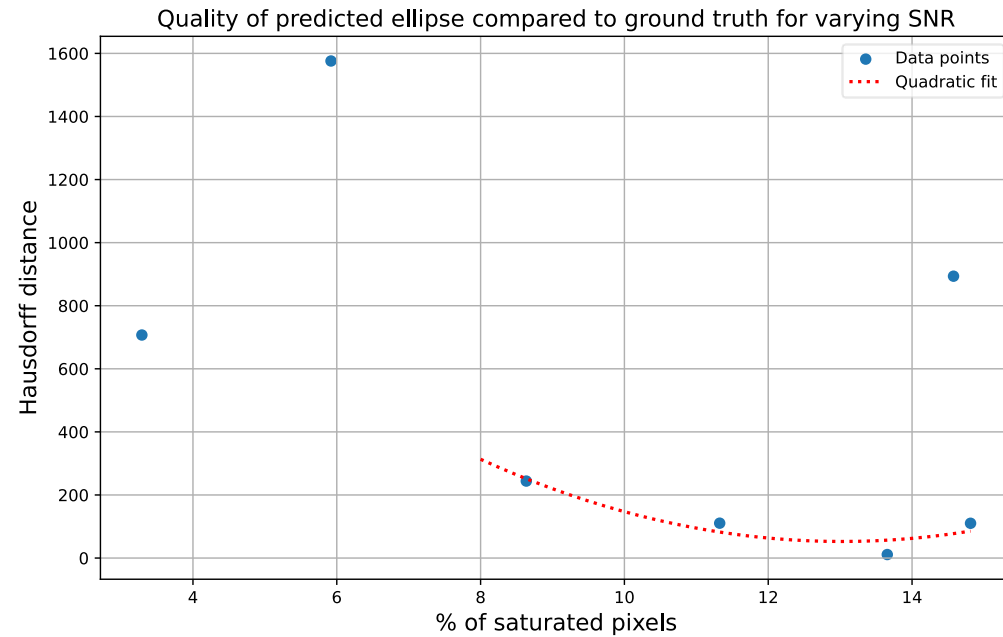
$$d_H = 1021.2, IoU = 0.620$$

Ellipse fit under variable SNR

WHEEL SINKAGE



- JAXA data

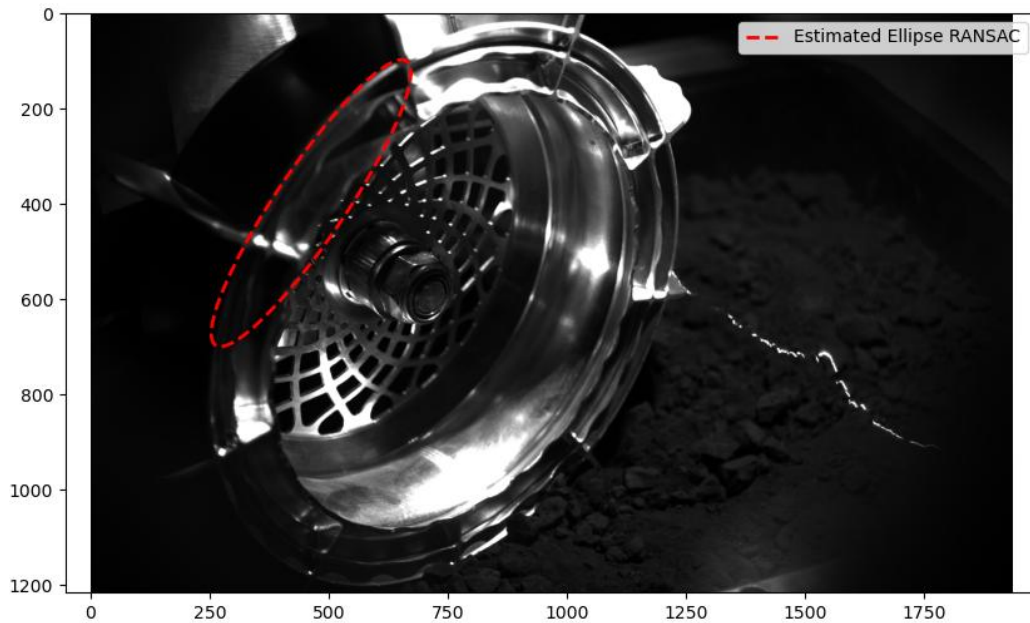


Ellipse fit under variable SNR

WHEEL SINKAGE

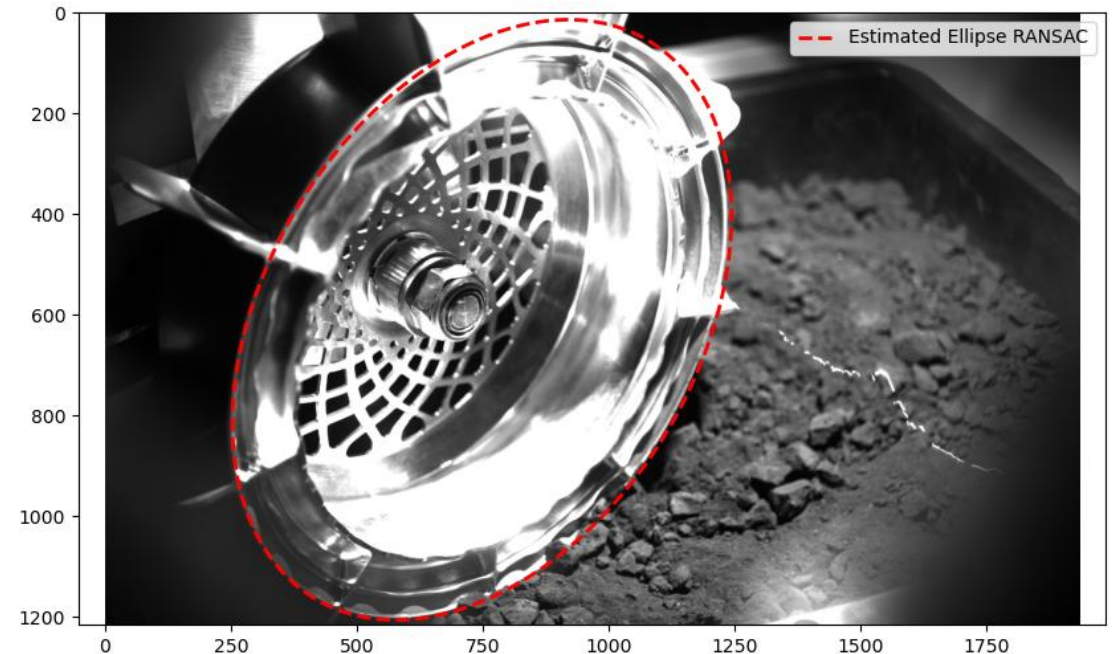


- JAXA data



Under-exposure, JAXA regolith simulant (3.29% saturation)

$$d_H = 707.12, IoU = 0.093$$



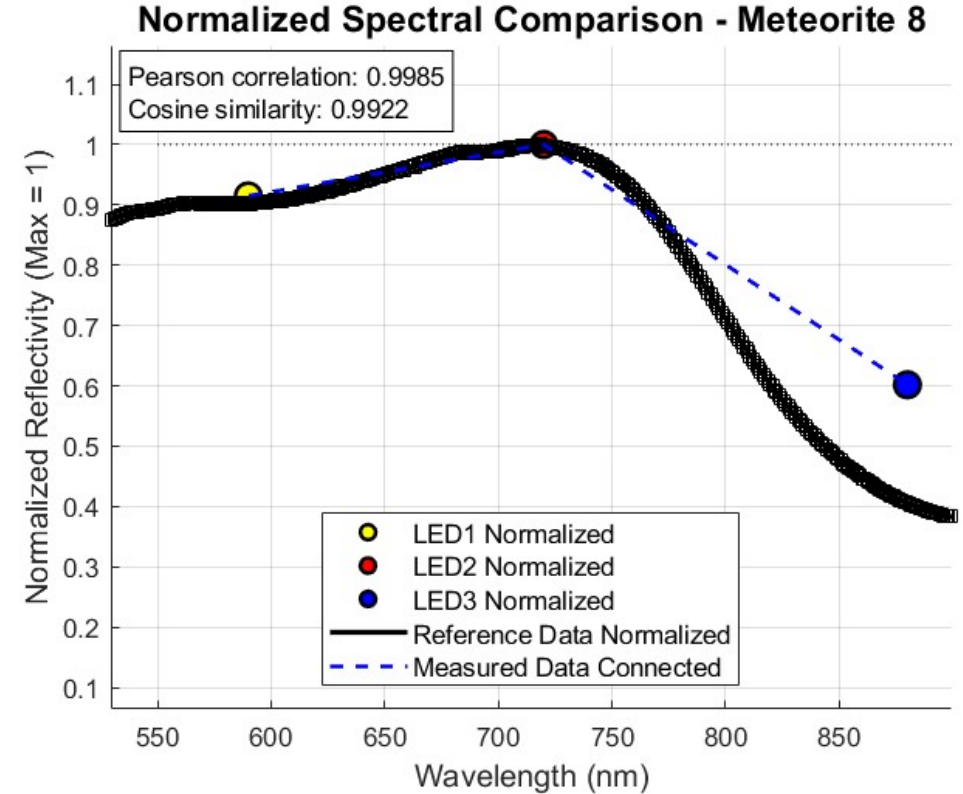
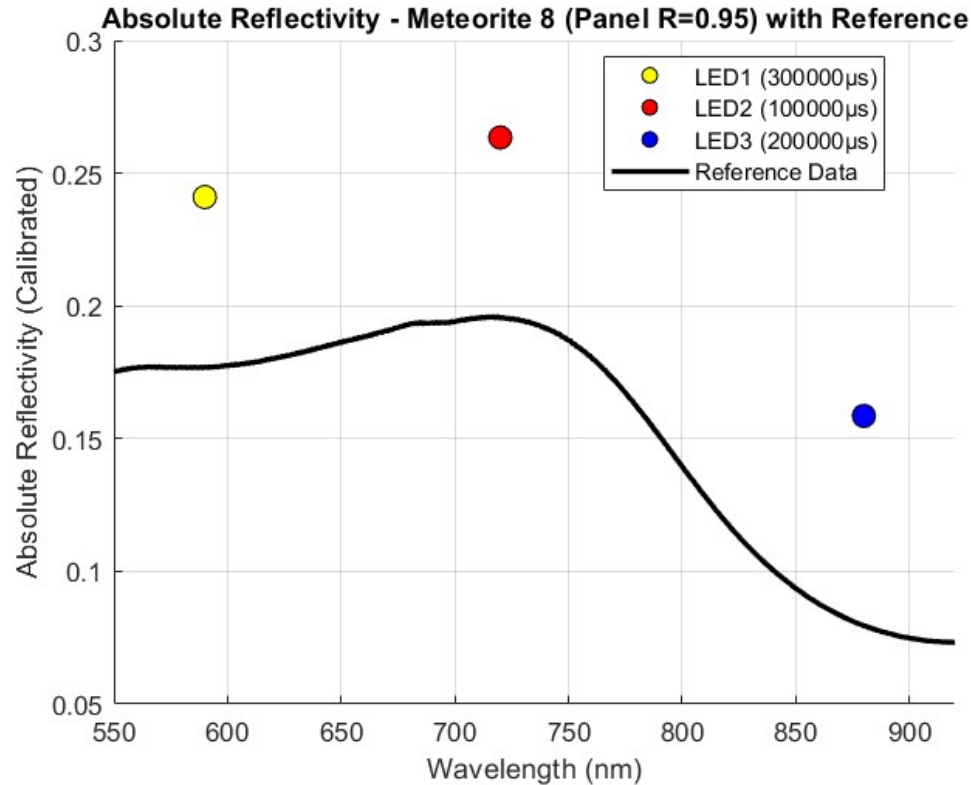
Optimal exposure, JAXA regolith simulant (13.6% saturation)

$$d_H = 11.092, IoU = 0.982$$



SPECTROSCOPY

METHODOLOGY AND RESULTS



PEARSON CORRELATION

$$r = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^n (x_i - \bar{x})^2} \sqrt{\sum_{i=1}^n (y_i - \bar{y})^2}}$$

COSINE SIMILARITY

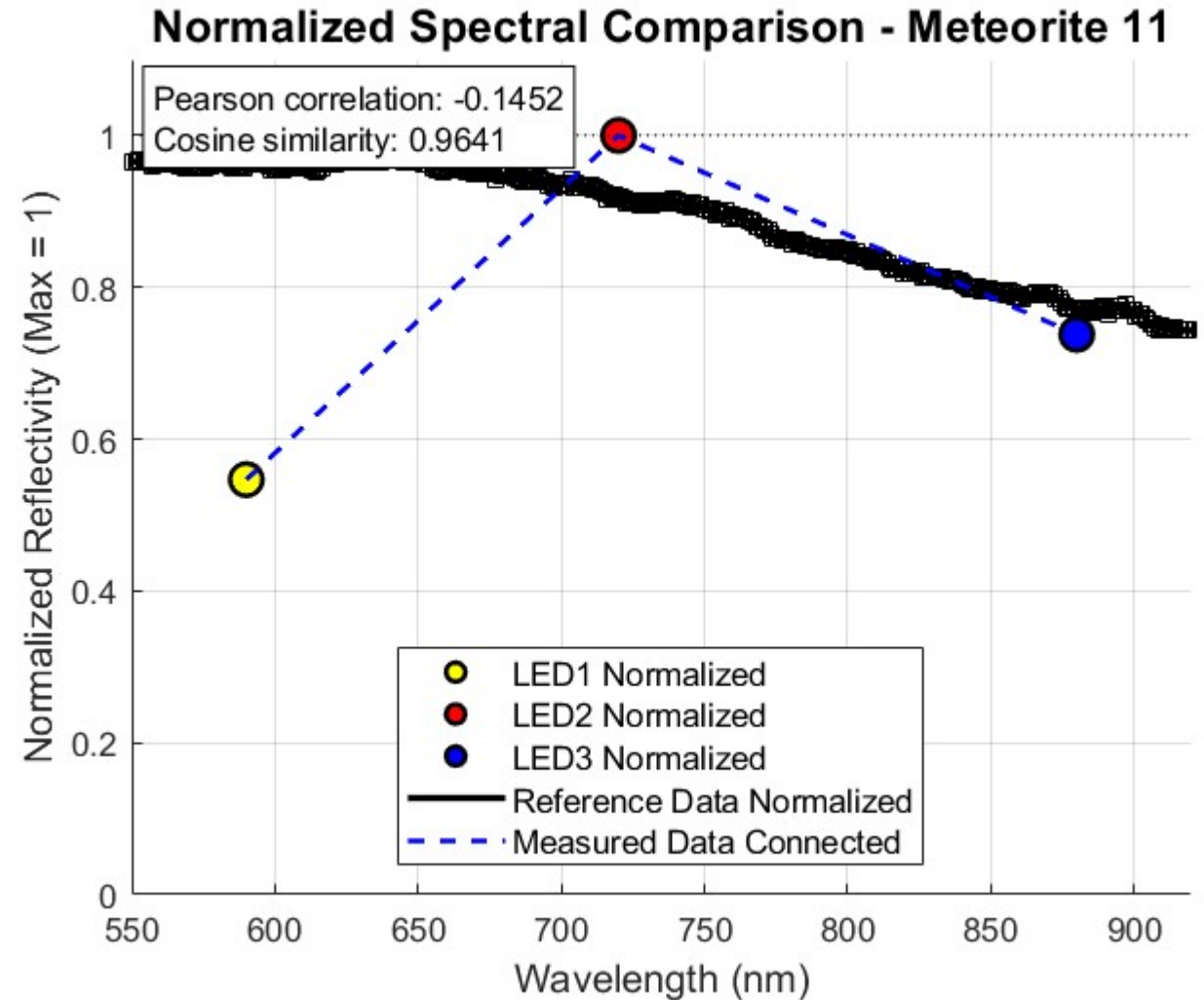
$$\text{CosSim} = \frac{\vec{x} \cdot \vec{y}}{\|\vec{x}\| \|\vec{y}\|} = \frac{\sum_{i=1}^n x_i y_i}{\sqrt{\sum_{i=1}^n x_i^2} \sqrt{\sum_{i=1}^n y_i^2}}$$

RESULTS

LAST SEMESTER

Paper	Pearson	Cosine
Blue	0.9869	0.9953
Dark Blue	0.9979	0.9987
Dark Green	0.9972	0.9986
Dark Pink	0.9913	0.9960
Green	0.9906	0.9956
Orange	0.9879	0.99498
Pink	0.9868	0.9949
Purple	0.9919	0.9967
Red	0.9949	0.9977
White	0.9843	0.9932
Yellow	0.9864	0.9939

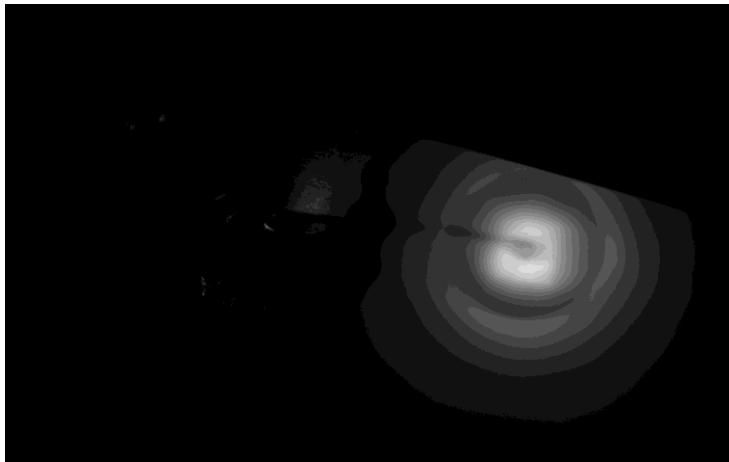
Meteorite	Pearson	Cosine
M3	0.8799	0.9975
M5	0.8066	0.9969
M6	0.9987	0.9994
M8	0.9986	0.9946
M9	0.8369	0.9875
M10	-0.2086	0.9849
M11	-0.0273	0.9760
M12	0.6263	0.9938



RESULTS

ANALOGUE MATERIALS

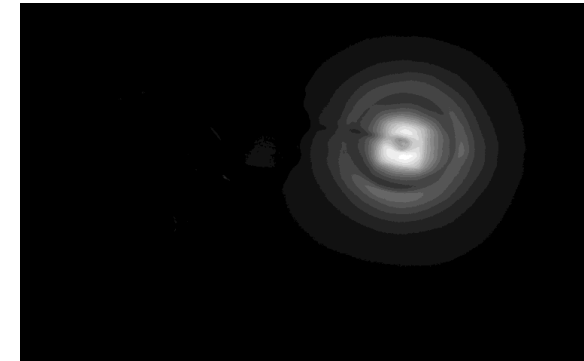
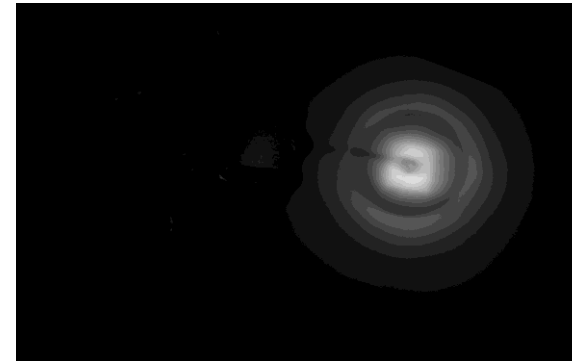
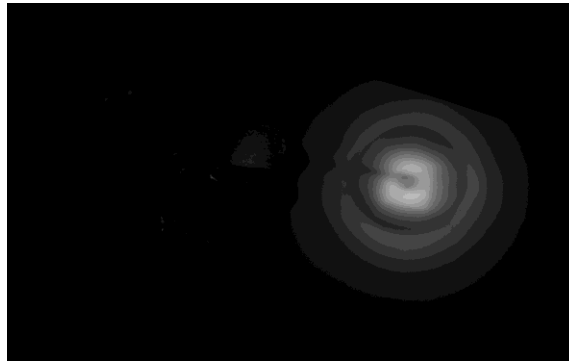
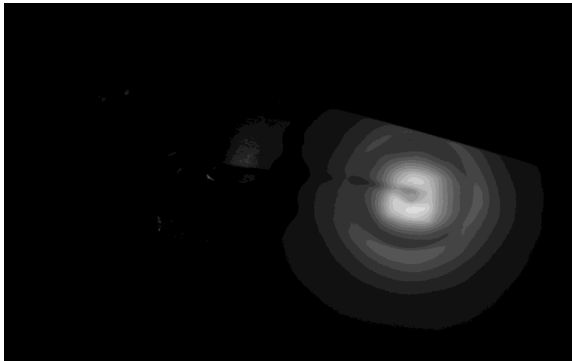
Point	Reference	Pearson	Cosine
Point 1	Olivine Model (USGS)	0.8361	0.9642
	Terrestrial Sample Olivine (CNRS)	0.9284	0.9875
Point 2	Olivine Model (USGS)	0.5579	0.8024
	Terrestrial Sample Olivine (CNRS)	0.7118	0.8529

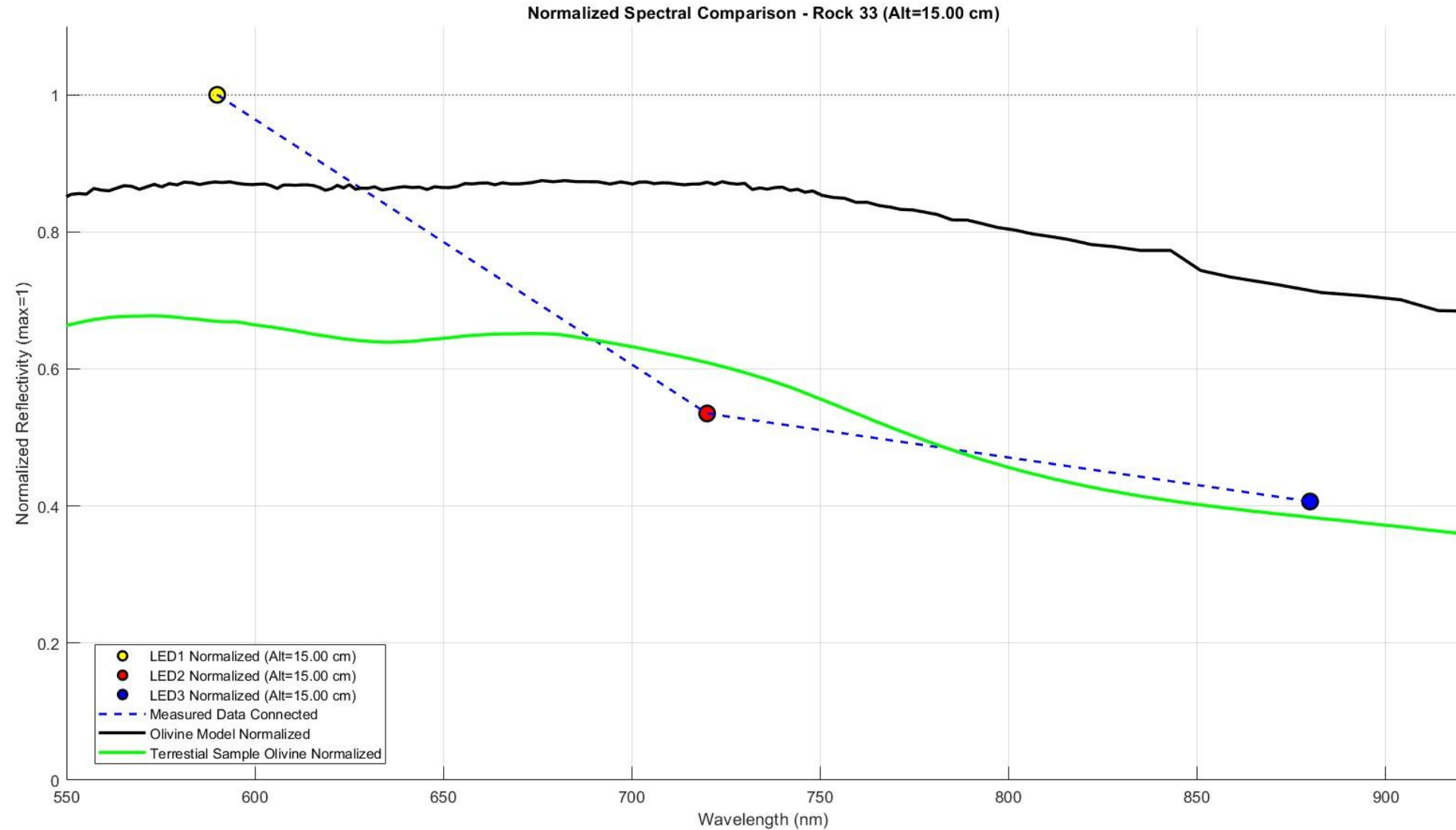


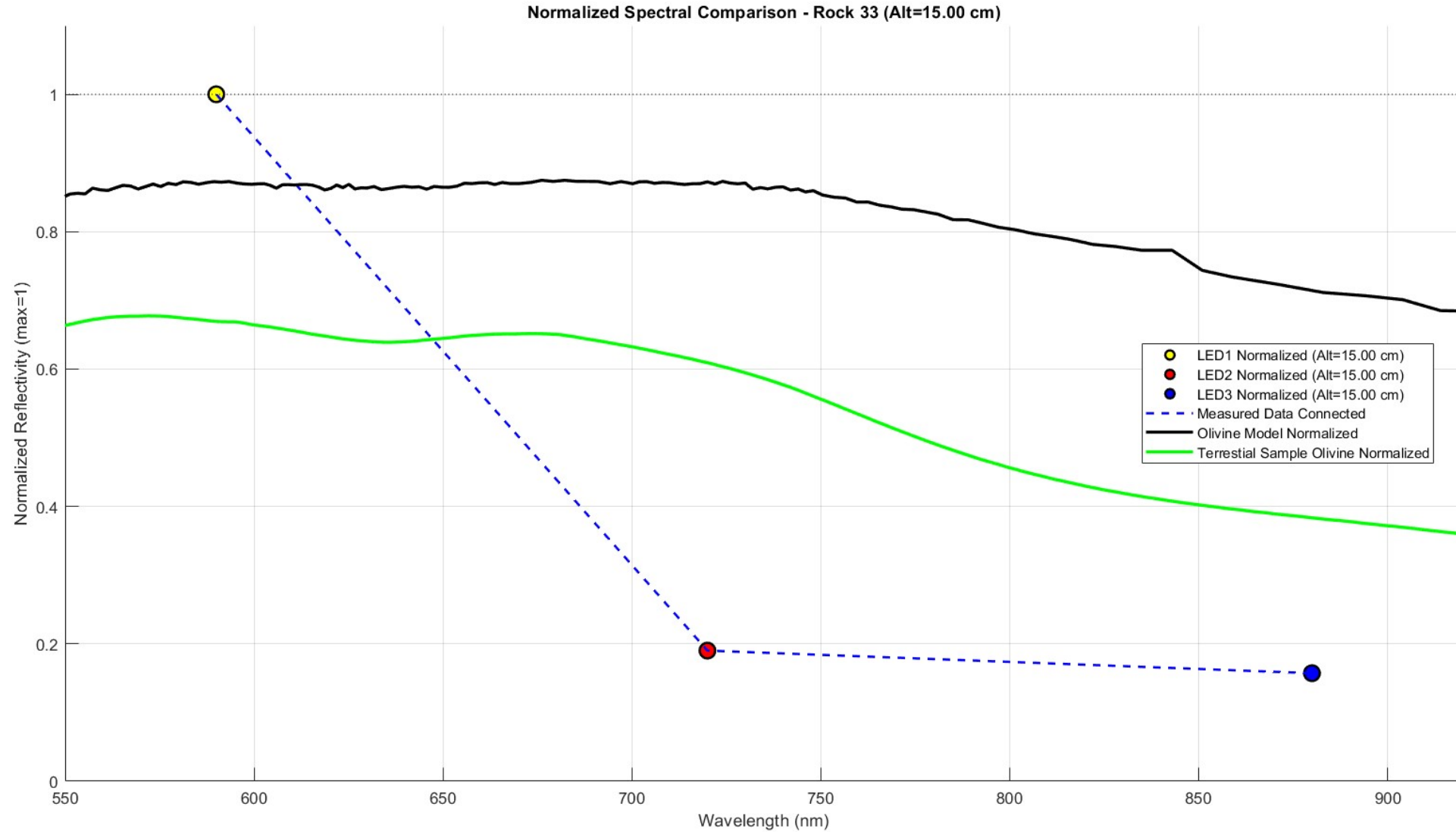
	Pearson	Cosine
(USGS)	0.9495	0.9918
le Pyroxene (CNRS)	0.1413	0.9675
(USGS)	0.9913	0.9473
le Pyroxene (CNRS)	0.3230	0.8960

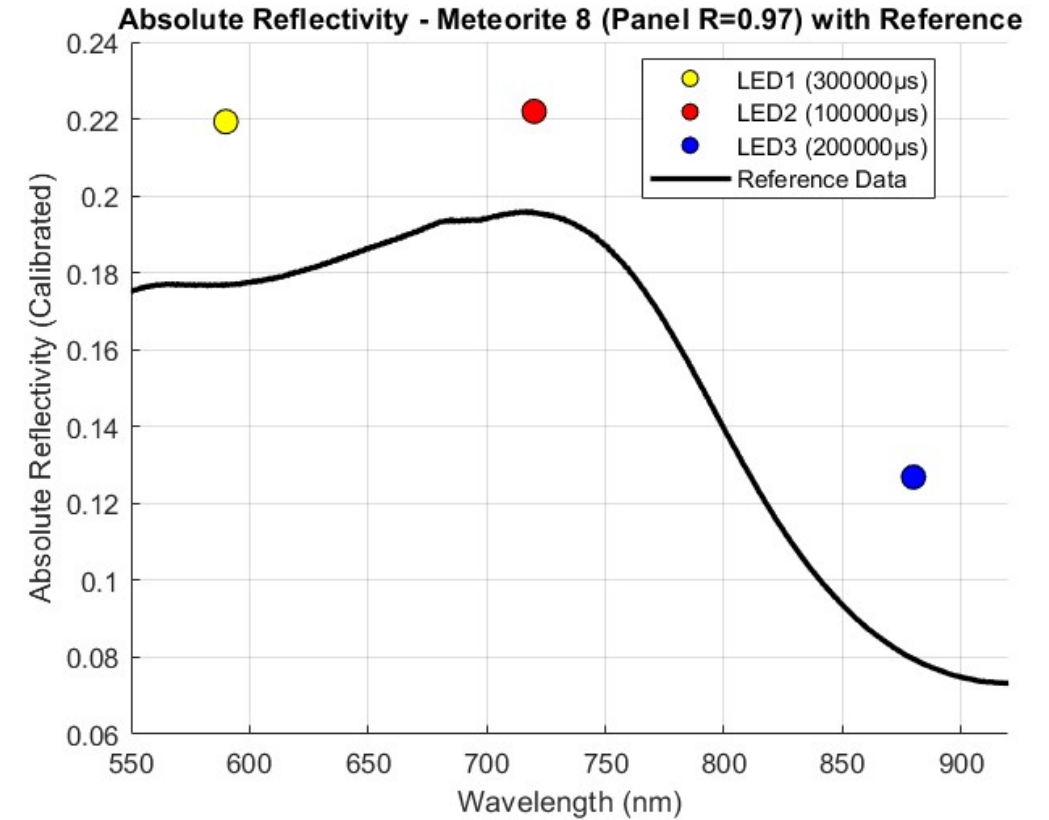
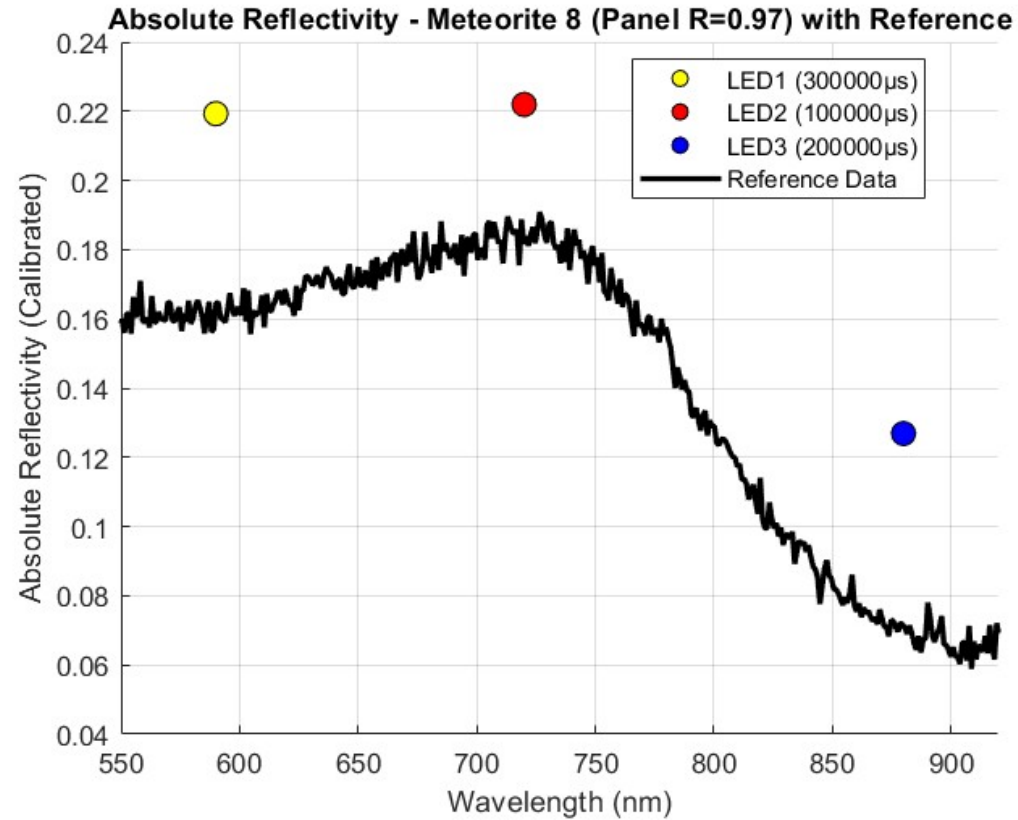
RESULTS

ANALOGUE MATERIALS



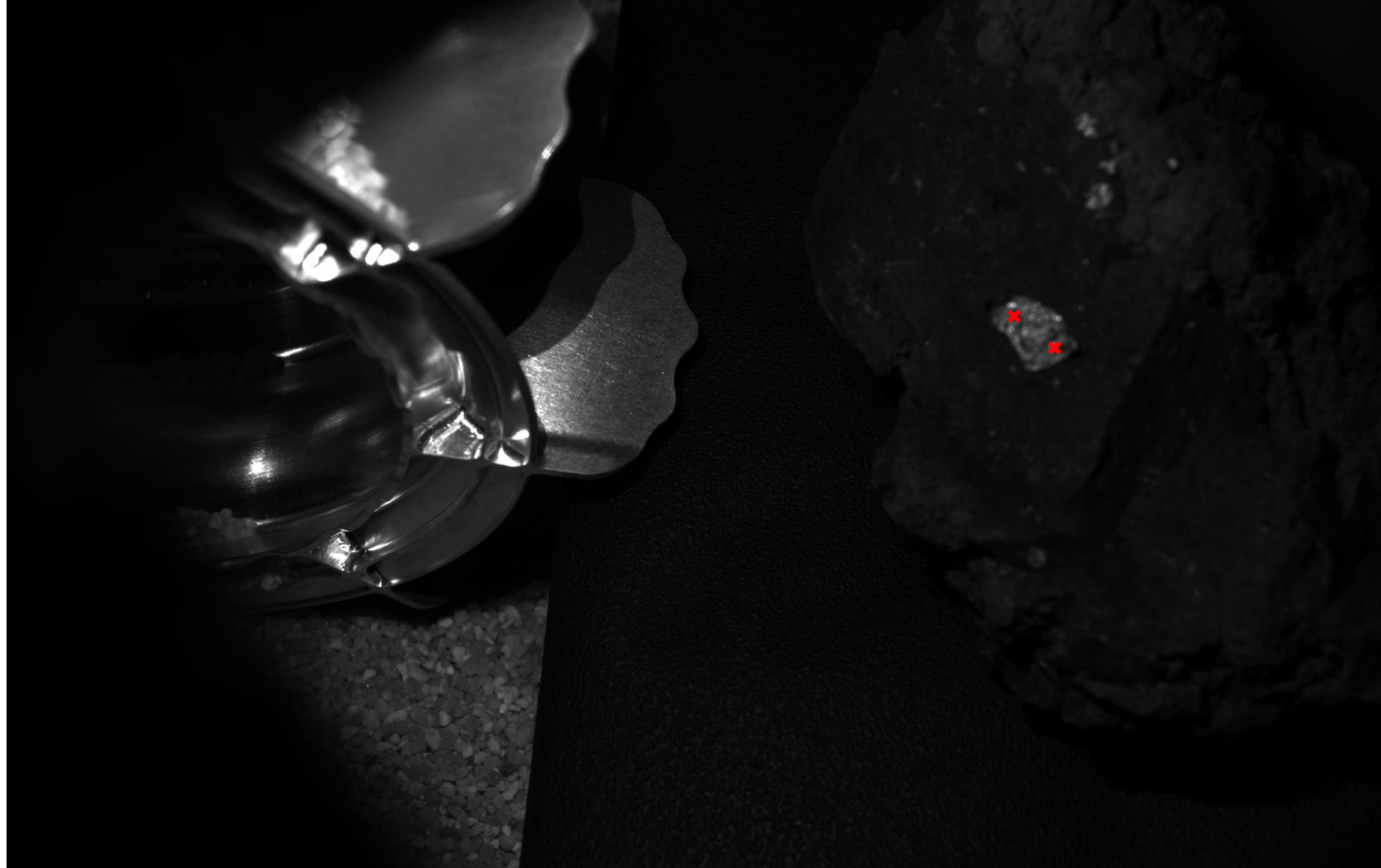








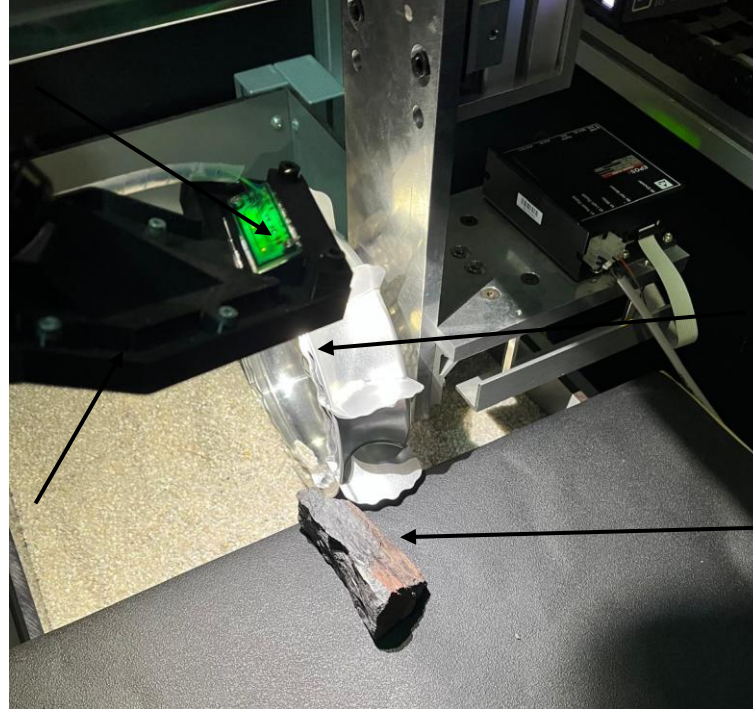
RESULTS



RESULTS

Testbed LEDs

*Testbed
WheelCam*



Testbed Wheel

Sample